MIL-R-38370B(USAF)
19 April 1972
Superseding
MIL-R-38370A(USAF)
16 December 1968

MILITARY SPECIFICATION

RECORDING SET, FLIGHT DATA A/A24U-10

1. SCOPE

1.1 This specification covers a flight data (VGH) recording set, designated A/A24U-10.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-R-38371	Reproducer, Flight Data MXU-386/E

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MIL-STD-721	Definitions of Effectiveness for Reliability, Maintaintability, Human Factors, and Safety
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The recording set shall consist of the following major components:

Component	See Requirement
MXK-383/A24U-10 Recorder	3.6.1
MXK-385/A24U Magnetic Tape Magazine	3.6.2
TRK-77/A24U Transmitter Accelerometer	3.6.3.

3.3 General specification. The requirements of MIL-E-5400 as specified for class 2 equipment apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3.1 Operating temperature. The maximum operating temperature shall be 71°C.

3.3.2 Altitude. The recording set shall operate as specified herein up to an altitude of 80,000 feet.

3.3.3 Input power. The input power shall be 28V dc as defined in MIL-STD-704, table titled "Steady-state dc voltage limits under different modes of electric system operation," equipment category B, except the recording set need not meet this requirement during the emergency conditions or during intervals when transient conditions cause input voltage to fall outside the 21 to 29V limit. The recording set shall be designed to withstand, without damage, the 80V transient conditions defined in MIL-STD-704.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.3.3.1 Power consumption. Total power consumption shall not exceed 50 watts.

3.3.4 Test points. The recording set test points shall be wired to a standard test connector as shown on figure 1 to permit rapid flight-line testing of the recording set by use of a TTU-226/E test set conforming to Conrac part No. 7950788, or equivalent, and for fault isolation of the recording set to a module level to the maximum practical extent.

3.3.5 Environmental conditions. Environmental conditions specified herein shall be in accordance with MIL-STD-810 except during the playback mode of operation, the tape magazine environment shall be in accordance with MIL-R-38371.

3.4 Design and construction. The recording set shall be designed to sense, measure, reduce, and record airspeed, vertical acceleration at the center of gravity, altitude, and elapsed time. When coupled with a suitable external sensor (see 6.2), the recording set shall also be capable of recording a reserve word simultaneously with the recording of the specified parameters. At the conclusion of the flight and with the aircraft on the ground, the tape magazine containing the recorded data shall be easily detached from the recording set. It shall be possible to attach the tape magazine to a reproducer conforming to MIL-R-38371 for reproduction of the data onto standard IBM, or equal, compatible tape reels for reduction by commercial data-processing equipment.

3.4.1 Approved design. The design used in the fabrication of the recording set shall be approved by the procuring activity. When specified (see 6.2), the contractor shall prepare design data to be submitted with or prior to the test outline specified in 4.4.3.1.

3.4.2 Data loss. The recording set shall be so designed that data lost or erroneous data gained (bit dropout or bit dropin) shall not exceed two bits per million.

3.4.3 Accuracy

3.4.3.1 Indicated airspeed, acceleration, and altitude. The total error of the combined recording set and reproducer caused by adding or interacting parameters of linearity, hysteresis, friction, repeatability, noise, environmental effects (including temperature), and any other causes of error attributable to the recording set and reproducer shall not exceed ±2 percent of full scale of the acceleration, altitude, and airspeed ranges defined in 3.6.1.1 and 3.6.1.3.

3.4.3.2 Elapsed time. The total error, including all accumulative errors, due to the internal clock, data handling, recording, and playback shall not exceed ±10 seconds or 2 percent of elapsed time for a 40-hour period of total elapsed time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.4.4 Reliability program. When specified (see 6.2), a reliability assurance program shall be established and reports prepared thereon in accordance with MIL-STD-785.

3.4.4.1 Reliability. The recording set shall have a minimum acceptable mean-time-between-failures (MTBF) of 200 hours.

3.4.5 Longevity. The recording set shall have an operating life span (equipment longevity as defined in MIL-STD-781) of 1,000 hours.

3.4.6 Maintainability. When specified (see 6.2), a maintainability program shall be established and reports prepared thereon in accordance with MIL-STD-470.

3.5 Performance. The recording set shall meet the requirements specified herein under the following conditions:

a. Temperature shock from -54° to +71°C

b. Applied acceleration of 15g along any axis

c. Temperature-altitude as specified for class 2 equipment (MIL-STD-810) at a maximum temperature of +71°C and a maximum altitude of 80,000 feet

d. Vibration as specified for catagory (b) equipment (MIL-STD-810) at room temperature, -54°, and +71°C

e. Shock of 15g along any axis

f. Salt laden atmosphere

g. Dust particles as encountered in desert areas

h. Operation in explosive atmosphere

i. Fungus growth as encountered in tropical climates.

3.5.1 Electromagnetic interference. The recording set shall meet the applicable electromagnetic interference requirements of MIL-STD-826.

3.6 Details of components

3.6.1 Recorder. The MXK-383/A24U-10 recorder and the MXK-385/A24U tape magazine may be mounted in any attitude at any location in the aircraft fuselage. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

recorder shall consist of the following major modules of which the digital computer shall be an interchangeable, easily replaceable, plug-in module:

a. Digital computer module

b. Power supply

c. Absolute pressure transducer module (altitude)

d. Differential pressure transducer module (airspeed)

e. Chassis.

3.6.1.1 Digital computer module. The digital computer module shall perform the following:

a. Convert analog acceleration data into nineteen discrete levels as specified in table I

b. Determine and count acceleration peaks (see 6.3.1) with a counting capacity for each of 16 levels (except levels A, B, and C) as specified in table I

c. Count total elapsed time in increments of 10 seconds for periods of 10 minutes as a function of a self-contained clock

d. Convert airspeed data and altitude data into discrete levels which form two 4-bit digital words, one for airspeed and another for altitude as specified in table II

e. Provide for storage of a 7-bit digital input word (reverse word). Reverse-word inputs to the recording set shall be as illustrated on figure 2 and shall appear at J3, Bendix connector PTO2P-14-19P, or equal. The following characteristics shall be displayed:

Reverse Word Signal Requirements

(1) Ground return, 1.0 ma maximum

(2) +4V digital logic gate control (0 ma)

(3) Logic 0, 0 ±0.2V dc
Logic 1, +4 ±0.2V dc

(4) Logic 0 or logic 1 noise levels shall not exceed ±0.2V dc

(5) Input capacitance shall not exceed 5 uuf

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:29Z
Check the source to verify that this is the currentversion before use.

f. Operate and provide all accumulated data to the tape magazine each time any one of the following conditions occurs:

(1) When any one of the acceleration peak counters has reached its capacity as specified in table I

(2) When the time word counter attains a count of 60 (equivalent to an elapsed time of 10 minutes)

Table I. Acceleration Level

Range (g) -3 to +9	Level Number	Counter Distribution	Max Decimal Count	System Frequency Response (Hz)
-1.8	1	1	1	1
-1.0	2	2	3	3
-0.6	3	2	3	3
-0.2	4	3	7	7
0.2	5	4	15	12
0.6	6	5	31	12
(0.8)	A			
(0.8)	B			
(1.2)	C			
1.4	7	5	31	12
1.8	8	4	15	12
2.2	9	4	15	12
2.6	10	4	15	12
3.0	11	3	7	7
3.8	12	2	3	3
4.6	13	2	3	3
5.4	14	2	3	3
6.6	15	2	3	3
7.8	16	2	3	3

(3) When a change in the indicated airspeed or altitude level occurs

(4) When the remote switch (see 3.6.4) is turned off

(5) When power is applied to the recording set either at the beginning of a flight or after the primary power has been interrupted

g. Program the record cycle by a self-contained serial programmer. The tape magazine shall record and run only during the record cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE II. Altitude and Indicated Airspeed Levels with Binary and Decimal Equivalents

ALTITUDE (1,000 Feet)	BINARY	DECIMAL	IAS (KNOTS)	BINARY	DECIMAL
0 to 1	1011	11	0 to 100	0000	0
1 to 2	1010	10	100 to 150	0001	1
2 to 5	1001	9	150 to 200	0010	2
5 to 10	1000	8	200 to 250	0011	3
10 to 15	0111	7	250 to 300	0100	4
15 to 20	0110	6	300 to 350	0101	5
20 to 30	0101	5	350 to 400	0110	6
30 to 40	0100	4	400 to 450	0111	7
40 to 50	0011	3	450 to 500	1000	8
50 to 60	0010	2	500 to 550	1001	9
60 to 80	0001	1	550 to 625	1010	10
			625 to 700	1011	11
			700 to 800	1100	12

3.6.1.1.1 Acceleration data channel frequency response during system operation shall conform to table I.

3.6.1.1.2 The digital computer module shall contain the necessary logic circuitry to count peak accelerations as defined under 6.3.1 and admit these counts into appropriate individual peak-level binary counters. Upon initiation of a record cycle, this data and the time counter shall be simultaneously transferred into a shift register and then immediately reset on the peak level counters. The time counter shall be reset only as a function of its own count of 60 after initiating a record cycle and transferring its data into the shift register. The altitude and indicated airspeed words shall be transferred into the shift register after the record cycle has been completed in preparation for the next record cycle.

3.6.1.1.3 The shift register shall be so arranged that when pulsed for a serial output, it will present its data to the record amplifier in such a manner that the data frame format will be as specified in table III which is to be read from left to right (the most significant binary bit first). A second channel shall be recorded for data bit definition and location. The binary status of each bit shall be as shown on figure 3. Both recording heads of the tape magazine shall be provided with the negative voltages illustrated on figure 3 during the nonrecording intervals which occur between frames of the data channels and the reference channels. During the prerecord-erase and the reproduce-erase modes, both heads shall record the negative voltages as shown on figure 3.

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:29Z
Check the source to verify that this is the currentversion before use.

TABLE III. Shift Register Capacity and Arrangement

Parameter	BOF	Reserve Word	Elapsed Time															IAS	ALT	Code Bit
Count Level Number			10 Sec Count	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
No. of Bits	1	7	6	1	2	3	4	5	5	5	4	4	4	3	2	2	2	2	2	4	4	1

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.1.1.4 Programmer. The programmer shall perform the following functions associated with a record cycle:

a. The sequency of events of a record cycle shall be as follows:

(1) Start the tape magazine

(2) Transfer all bits of data accumulated in the counters, plus the digital words and one coding bit, simultaneously into the shift register

(3) Reset all counters bits except the time period counter

(4) Fifty milliseconds after the start of the recording media, begin shifting out the data contained in the shift register in serial form for the nonreturn-to-zero (NRZ) recording mode. Provide reference pulses on a second recording channel for data bit definition that are synchronized with the bits of serial data on the data recording channel as shown on figure 3

(5) Upon completion of pulsing out the shift register, stop the tape magazine

(6) Cease to provide shift pulses to the shift register and reference channel

(7) Transfer the two 4-bit words (altitude and indicated airspeed) into the shift register.

b. The maximum rate for successive printouts shall be 1 second. The readout cycle shall be initiated for any of the following occurences:

(1) A maximum capacity condition of any of the peak acceleration level counters

(2) A count of 60 (minutes) in the time-period counter

(3) Any change of digital data on either of the two 4-bit words (altitude and indicated airspeed)

(4) When power to the set is first applied, a beginning-of-flight (BOF) bit shall be generated and recorded, thus marking the first data frame recorded for this initial power-on condition

(5) A delay in power to the set shall be provided to permanently record that data contained in the binary counters when a remote cockpit switch is turned off.

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:29Z
Check the source to verify that this is the currentversion before use.

3.6.1.2 Power supply. The power supply in the recorder shall be gasket sealed. With input power as specified in 3.3.3, the power supply shall be capable of supplying all the recording-set power and reference voltages. Safety provisions shall be incorporated into the power supply to protect against transient conditions and to insure that the output will not damage the recording-set circuitry under overload, short-circuit, open-circuit, or under adverse environmental conditions.

3.6.1.3 Pressure transducers. The recorder shall contain two pressure transducers. Each transducer shall generate data, one corresponding to altitude and the other corresponding to indicated airspeed information. The output of the transducers shall be scaled so that the computer module can convert this data into two separate 4-bit binary words. The correspondence between computer words and flight data levels is shown in table II.

3.6.1.3.1 Range. The range shall be as follows. The altitude and indicated airspeed values shall take into account the effect of the standard atmosphere specified in MIL-STD-859.

a. Absolute transducer - 0 to 80,000-foot altitude

b. Differential transducer - 0 to 800 knots indicated airspeed.

3.6.1.3.2 Overpressure. The transducers shall withstand a 25-percent overpressure with no damage or degradation of performance.

3.6.1.3.3 Transducer volume. The case volume of the pressure element enclosure for each transducer shall not exceed 55 cc.

3.6.1.3.4 Accuracy. The accuracy of the pressure transducers shall be such that the total error, including errors due to the transducers, digital computer and record and playback system shall not exceed ±2 percent of full scale as defined in table II.

3.6.1.4 Recorder chassis. The recorder chassis shall have provisions for plug-in connections and for secure attachment of all modules. The chassis shall be mounted on vibration isolators to protect the recorder against shock and vibration. The complete recorder, with isolators, shall remain within specified dimensions. The lowest resonant frequency of the computer-recorder, with isolators, shall be greater than 13 Hz. The isolators shall withstand repeated loads of 20g in any axis at the equipment center-of-gravity without reaching maximum deflection.

3.6.1.4.1 Outline dimensions. The recorder outline dimensions shall be as shown on figure 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.1.4.2 Shock mounts. The computer-recorder shock mounts shall be designed to withstand repeated loads of 20g in any axis without reaching maximum deflection and to meet the vibration requirements specified herein.

3.6.1.4.3 Mounting plate. The computer-recorder mounting plate shall be as shown on figure 4.

3.6.1.5 Recorder reliability. The recorder shall have a minimum acceptable MTBF of 300 hours.

3.6.2 Tape magazine MXK-385/A24U. The MXK-385/A24U tape magazine shall record and play back digital information on two channels in the NRZ mode in frames of fixed word lengths. In the record mode, a signal shall start the tape and after a 50-msec delay, the digital information shall be recorded. Upon completion of the recording of the frame, a signal shall stop the tape. This mode may continue until the storage capacity of the tape (3,600 data frames minimum) is reached. For the playback mode, the recorder magazine may be removed from the recording set and connected onto a ground reproducer. The recorder information may then be transferred to other storage media and after data verification, the tape may be erased.

3.6.2.1 Recording method. Digital information shall be recorded on one of the two channels with NRZ system. The other channel shall contain clock reference pulses. The information bits shall coincide with the reference pulses for bit definition. The precise form of recording and playback voltage shall be as shown on figure 3.

3.6.2.2 Packing density. The packing density shall be as least 200 bits per inch of tape.

3.6.2.3 Information frame. Each information frame to be recorded shall be a fixed length. The total number of bits per frame shall be 70. Similarly, the clock reference channel shall contain 70 reference pulses corresponding to the 70 bits of information contained in the information channel.

3.6.2.4 Tape speeds. During the record mode, the tape speed shall be nominally 15/32 inch per second and during the playback mode, tape speed shall be nominally 7.5 inches per second. Tape speeds in either record or playback modes of operation shall vary no more than ±10 percent of nominal values under normal conditions. An additional ±15 percent variation will be permitted during adverse combinations of environmental extremes.

3.6.2.5 Start-stop time. During the record mode, the tape start time shall not exceed 60 msec, maximum, and the stop time shall not exceed 100 msec, maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.2.6 Magazine control. Tape start-stop shall be controlled by command of computer-recorder signals. Application of a signal shall start the tape and removal of the signal shall stop the tape. Figure 5 illustrates the electrical interface between the tape magazine and computer-recorder.

3.6.2.7 Tape capacity. Total tape capacity shall be 160 feet in a continuous loop. A power interlock shall be provided at the beginning of the tape to prevent recording over data already recorded. An indicator of 99 percent of tape remaining shall be synchronized with the interlock.

3.6.2.8 Crosstalk and noise. The magnitude of electromagnetic or electrostatic or any other form of crosstalk or noise shall be 30 db below the nominal output level of the affected channel when reproduced by the reproducer.

3.6.2.9 Playback voltage. Playback voltage shall be a minimum of 15 mv peak-to-peak at a playback speed of 7.5 inches per second. Playback voltage shall never be less than 7 mv peak-to-peak under all life and environmental record conditions.

3.6.2.10 Tape magazine reliability. The tape magazine shall have a minimum acceptable mean-cycles-between-failures (MCBF) of 25. Reliability and longevity environmental conditions shall be as specified under 4.6.15.

3.6.2.11 Tape magazine longevity. The tape magazine longevity shall be 90,000 cycles of frame recording, i.e., tape start, record, tape stop, and for 25 cycles of recording and playback; i.e., detaching the tape magazine from the recording system, attaching the tape magazine to the playback system, playback of recorded data, and erasing the recorded data. With repair and maintenance, the operating life of the magazine shall be one-million cycles of frame recording and 250 cycles of recording, playback, and erasing.

3.6.2.12 Outline dimensions for the tape magazine shall be as shown on figure 6.

3.6.2.13 Mounting. The magazine shall be designed for easy accessibility with secure and positive mounting to the recorder and the reproducer in the same manner. The tape magazine shall contain no protruding elements, such as mechanical and electrical fittings. All mechanical and electrical fittings shall be contained in recessed areas. The tape magazine shall be a plug-in unit that can be easily mounted to and removed from the computer-recorder package without removing the computer-recorder from the aircraft.

3.6.2.14 Tape-remaining indicator. A visual indicator of percentage of tape remaining shall be provided as an integral part of the tape magazine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.2.15 Visual indicator. A visual indicator shall be provided as an integral part of the tape magazine and shall operate when a data word is being recorded.

3.6.2.16 Record playback heads. The recorder tape magazine shall contain two similar dual-channel record-reproduce heads. Separation of the heads, measured between gap centerlines, shall be not less than 1 inch. In the record mode, the tape shall pass over the first head where a direct-current erase on both channels is performed and then over the record head where data and clock reference channels shall be reproduced by the second head with the first interlocked by external circuitry to prevent accidental overwrite or erase. Track width of the erase and record heads shall be 0.088 inch and 0.076 inch, respectively.

3.6.2.17 Erase head. The magazine shall be so designed that the erase head will always be energized by the computer-recorder in the record mode.

3.6.2.18 Skew between channels. The maximum skew shall not exceed ±10 percent between corresponding pulses of the reference and data channels during the playback speed of 7.5 IPS. Corresponding pulses are those bits recorded on the reference channel used for bit definition of the data channel. The skew shall be measured on the leading edges of the corresponding pulses.

3.6.2.19 Shelf life. The tape magazine shall exhibit no degraded performance after a minimum shelf life of 1 year.

3.6.2.20 Tape reel brake. A positive locking device shall be provided for the tape reel to prevent motion due to possible angular acceleration during transportation of the tape magazine. The reel brake shall provide for a power interlock to prevent tape magazine operation when the brake is engaged. The tape reel brake shall be engaged when electrical interlock is activated and disengaged when the interlock is defeated. This brake shall be released by engaging a dowel into the proper recessed mounting hole of the tape magazine (see figure 6). Under no conditions shall the tape advance when the mechanical braking mechanism is engaged.

3.6.2.21 Tape transport case. The tape transport case shall be so designed that it can be used to ship tape magazines by U. S. mail or other common carriers without damaging the magazine. Magnetic shielding shall be provided to prevent electrical erasure of the tape by strong magnetic fields during shipment.

3.6.2.22 Tape magazine electrical interface. Pin designations and tape magazine functions shall be as shown on figure 5.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.2.23 Tape magazine and tape transport case quantities. Unless otherwise specified (see 6.2), the recording set shall be provided with a tape magazine affixed to the computer-recorder and a second tape magazine, identical in every way and mounted in a tape transport case, shall also be provided. Each magazine shall have 99 percent of tape remaining.

3.6.2.24 Construction. The tape magazine shall be as simple and foolproof as possible as to operation, installation, and maintenance and shall not require alignment or recalibration during the life specified in 3.6.2.11. The magazine shall be enclosed in a hermetically sealed module and shall be completely interchangeable in all cases. The unit shall be so constructed that it can be directly attached to a reproducer conforming to MIL-R-38371 for the playback mode of operation. Record and playback amplifiers shall not be contained in the tape magazine.

3.6.3 Servo accelerometer. The TRK-77/A24U accelerometer shall be of the servo self-balancing type, shall have a range of -3 to +9g, and shall have a threshold value of -0.8 to +1.2g. Accelerometers shall be mechanically and electrically interchangeable so that any accelerometer may be used with any recording set without modification, calibration, or adjustment.

3.6.3.1 Wiring. The accelerometer shall be wired into the recording set by means of three shielded wires and shall be provided with an electrical connector equal to and interchangeable with Bendix pigmy connector PCIH-8-4P (100).

3.6.3.1.1 Accelerometer interface. The accelerometer interface shall be compatible with input requirements of the recorder. Electrical connection between the accelerometer and recorder shall be made by means of Bendix connector PT02P-12-10S, or equal. Pin designations shall be as shown on figure 7.

3.6.3.2 Scaling. The accelerometer output shall be direct-current voltage ranging from 0V at -3g to 7.028V at +9g. The maximum ripple shall not exceed 2 mv peak-to-peak exclusive of ambient vibration.

3.6.3.3 Frequency response. The accelerometer shall measure acceleration over a band width of 0 to 12 Hz.

3.6.3.3.1 Output. The accelerometer output shall be flat within ±1 percent at the upper frequency limit. At a frequency of 1.5 times the upper limit, the output shall be down at least 1.5 db and from that point on the frequency shall fall at a rate of at least 18 db per octave. The accelerometer output attenuation characteristics shall be at a minimum as specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

Table IV. Accelerometer Output Attenuation

Frequency (Hz)	Output Attenuation Ratio
6	1.0
7	1.0
8	1.0
9	1.0
10	1.0
11	1.0
12	1.0
13	0.99
15	0.976
20	0.944
25	0.50
30	0.28
40	0.125
50	0.064
60	0.036
70	0.022
80	0.015
90	0.01

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.3.3.2 Accuracy. The accuracy of the accelerometer shall be such that the error from acceleration input to voltage output shall not deviate by more than ±0.7 percent of full scale voltage, i.e., 7.028V from a straight line connecting the full scale output end-points from 0V to 7.028V. This percentage shall be the maximum permissable error detectable in the output of the accelerometer and shall include variations due to errors such as: Linearity, hysteresis, friction, repeatability, resolution, noise, and any other causes of error attributable to the accelerometer.

3.6.3.3.3 Temperature accuracy. An additional error of ±0.9 percent shall be permitted for ambient temperature. This error is defined as the difference between accelerometer output readings taken at room temperature and readings taken at any other temperature over a range of -54° to +71°C.

3.6.3.3.4 Source impedance. The source impedance of the accelerometer output shall be not greater than 100 ohms.

3.6.3.4 Load impedance. The load impedance of the circuit to which the accelerometer will be connected shall be 50,000 ohms ±0.5 percent.

3.6.3.5 Input power. Power to the accelerometer shall be disbributed through the recording set accelerometer connector. Input power shall be provided at a 28V d-c level as specified in MIL-STD-704, category B. No damage shall occur during the specified 80V d-c transient conditions.

3.6.3.6 Size and configuration. Size and configuration of the accelerometer shall be as shown on figure 8.

3.6.3.7 Weight. The total weight of the accelerometer shall not exceed 20 ounces.

3.6.3.8 Mounting. The accelerometer shall be supplied as a separate mountable component. The maximum outline of the accelerometer, its mounting holes, and surface shall be as shown on figure 8.

3.6.3.9 Orientation. The accelerometer shall indicate 1g under static conditions when placed on its base in a normal upright position.

3.6.3.10 Axis alignment. The alignment of the sensitive axis of the accelerometer shall be maintained 90° ±0.5 percent from the base or mounting bracket.

3.6.3.11 Overrange. The accelerometer shall not be damaged nor suffer degradation of performance when subjected to a steady-state acceleration of 20g along any axis in any direction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.6.3.12 Accelerometer reliability. The accelerometer shall have a minimum acceptable MTBF of 5,000 hours.

3.6.3.13 Longevity. The accelerometer shall have a minimum longevity of 10,000 hours.

3.6.4 Remote switch. A switch which can be mounted at a remote location shall control primary power to the system and shall be as shown on figure 9. Unless otherwise specified (see 6.2), the remote switch shall not be supplied as a part of the system.

3.6.4.1 Operating requirements. Electrical continuity between the remote switch and the recording set shall be made by means of the Bendix connector PTO2P-14-12P, or equal, as shown on figure 9. The switch shall be capable of continuous operation during 1.8-ampere 28V d-c load conditions. The remote switch shall operate without damage when subjected to the transient conditions outlined in MIL-STD-704.

3.7 Part numbering of interchangeable parts. All components having the same manufacturer's part number shall be functionally and dimensionally interchangeable as defined in MIL-STD-721. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.8 Electromagnetic interference suppression. The recording set shall meet the electromagnetic interference suppression requirements of MIL-STD-826 and shall operate satisfactorily in conjunction with other equipment which may be nearby. Recording set operation shall not be adversely affected by radio frequency voltages and fields reaching it from external sources nor shall the recording set itself be a source of interference for external equipment.

3.9 Dimensions. The overall dimensions of the recording set shall be in accordance with figures 4, 6, and 8.

3.10 Weight. The total weight of the recording set (recorder, tape magazine, and accelerometer) shall not exceed 15 pounds.

3.11 Operational markings

3.11.1 Accelerometers shall be clearly and permanently marked on the exterior of the case to show the location of the center of the seismic mass. In addition, an arrow shall be clearly and permanently marked on the case indicating the sensitive axis and which face should be up when normally installed in the aircraft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.12.1 Additional markings. The following additional markings shall be included:

a. Recorder
Type ___*_____
Serial No. __*_____

b. Tape Magazine
Date Installed _____ Date Removed _____

Recorder Serial No. _____ Range _____
Aircraft Type _____ Serial No. _____
Aircraft Station or Squadron _____
Remarks _____

c. Accelerometer
Type _____*_____ Range ____*_____
Serial No. _____

*Data to be entered by the contractor.

3.13 Workmanship. The recording set shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing, and freedom of parts from burrs and sharp edges.

3.13.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, of performance of the computer, they shall be held or limited accordingly.

3.13.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.13.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.13.4 Gears. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from backlash.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

3.13.5 Cleaning. The recording set shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash which might crumble shall also be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of recording sets shall be classified as follows:

a. Preproduction tests . . . . . Set 4.4

b. Quality conformance tests . . . . . See 4.5

4.3 Test conditions

4.3.1 Standard atmospheric conditions. When the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be conducted at atmospheric pressure (approximately 29.92 inches Hg), at room temperature (approximately 25°C), and a relative humidity of not greater than 90 percent. When tests are conducted with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the recording set shall be tested in its normal operating position.

4.3.3 Tapping. During static accuracy testing, the recording set shall not be tapped or vibrated prior to reading the outputs.

4.3.4 Test voltages. Unless otherwise specified, the recording set shall be tested with 28V dc. The voltage shall be maintained within 1V.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the current version before use.

4.3.5 Accuracy. Static accuracy tests shall be conducted with both increasing and decreasing pressures to permit adequate determination of hysteresis, drift, and after effect of both altitude and airspeed transducers. Static accuracy tests shall be performed on the accelerometer and recorder to insure that the specified performance at normal temperature is met.

4.3.5.1 Measurement tolerances shall be as follows:

a. Input
   (1) Static pressure (altitude) . . . . . 0.005 inch Hg
   (2) Impact pressure (IAS) . . . . . 0.005 inch Hg
   (3) Acceleration . . . . . ±0.1 percent full scale

b. Output
   (1) Digital readout . . . . . on-off lights/binary type writer printout
   (2) Other approved method.

4.3.6 External connections. When it is necessary to provide external connections to simulate installed conditions, they shall consist of standard connectors and at least 10 feet of copper or aluminum tubing attached to the pitot and static pressure ports of the recording set. The copper or aluminum tubing shall be coiled and arranged to permit drainage of condensed vapors.

4.3.7 Test equipment. During testing of the recording set, the pressure connections shall be connected to precision liquid barometers, pressure gages, or manometers the accuracy of which has been verified by comparison to secondary standards. If simulators are used to simulate accelerometer inputs, the accuracy of voltage simulation shall be ±0.1 percent.

4.3.8 Test setup. The recording set and the required test equipment shall be connected as specified in the applicable test. The following examination of the test setup shall be made and recorded with each new setup or after an off-time period:

a. Check zero adjustment of pressure measurement equipment

b. Check to see that the test equipment is functioning properly

c. Check to see that all electrical and pneumatic connections are completed and secured

d. Check the inspection tags of the test equipment to see that they are valid and not outdated.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the currentversion before use.

4.3.8.1 When changing from one chamber condition to another, the rate of temperature and pressure change shall not exceed 1°C per second and 0.5 inch Hg per second, respectively. The ambient temperature shall be controlled, as required, by means of thermocouples. During the high and low temperature tests, the recorder temperature shall be monitored at three locations as shown on figure 10. The thermocouple readings shall be recorded at the indicated time levels.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of five recording sets representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Test report. Upon completion of the preproduction tests, the contractor shall prepare a test report in accordance with MIL-STD-831 which shall contain sufficient data for subsequent analysis of test results and evaluation of equipment operation.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.4.3.1 Contractor-prepared test outline. When specified, (see 6.2), the contractor shall prepare an outline of tests and procedures to be used, including samples of methods of recording test data. Such outline shall include all tests, including those specified herein, necessary to determine compliance with the requirements of this specification. The test outline shall be submitted to the procuring activity for approval prior to formal testing. During the test program, if the test outline is found to be inadequate by the procuring activity it shall be amended as agreed upon between the contractor and the procuring activity.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests . . . . . See 4.5.1

b. Sampling tests . . . . . See 4.5.2.

4.5.1 Individual tests. Each recording set shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Pressure transducer accuracy tests . . . . . See 4.6.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

c. Acceleration data channel accuracy test . . . . . See 4.6.3

d. Reference time data channel test . . . . . See 4.6.4

e. Recording set frequency response . . . . . See 4.6.5

f. Recording set operation. . . . . See 4.6.6.

4.5.2 Sampling tests. Two recording sets shall be selected at random from each one hundred or less produced on the contract or order and subjected to the vibration tests specified under 4.6.7.

4.5.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests related to the failure shall be repeated.

4.5.2.1.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or items produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and methods of correcting them.

4.5.4 Environmental deterioration. Deterioration or corrosion of any internal or external part that might prevent continued safe or satisfactory operation of the recording set during its service life shall constitute failure.

4.6 Test methods

4.6.1 Examination of product. The recording set shall be visually inspected to determine that construction and finish are consistent with accepted standards for material and workmanship and to assure compliance with applicable specifications and standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.2 Pressure transducer accuracy. The recording set and its associated test equipment shall be set up, examined as specified under 4.3.8, and tested as follows:

4.6.2.1 Altitude transducer overpressure. The pitot and static pressure connections of the recording set shall be joined by means of a T-fitting. A pressure of 37.4 inches Hg shall be applied (this shall exclude sea-level pressure). The output shall indicate an overpressure from sea level to 37.4 inches Hg.

4.6.2.2 Indicated airspeed transducer overpressure. The static pressure connection shall be vented to atmosphere. A source of pressure shall be connected to the pitot pressure connection. A pressure of 53.75 inches Hg (gauge) shall be applied to the pitot connection. The output shall indicate an overpressure from 43.87 to 53.75 inches Hg gauge.

4.6.2.3 Altitude transducer leakage. The pitot and static pressure connections of the recording set shall be joined by means of a T-fitting. The T-fitting shall be connected to a differential pressure indicator and a source of vacuum. The high side of the pressure indicator shall be vented to the atmosphere. A differential pressure of 30 inches Hg shall be produced between the chamber and the atmosphere. When the specified differential pressure has been obtained, the source of vacuum shall be sealed off. The rate of pressure leakage shall e measured and shall not exceed 0.050 inch Hg per minute.

4.6.2.4 Indicated airspeed transducer leakage. The static pressure connection shall be vented to the atmosphere. A pressure indicator and a source of pressure shall be connected to the pitot pressure connection of the recording set. A pressure of 20 inches Hg (gauge) shall be applied to the pitot connection. When the specified pressure has been obtained, the source of pressure shall be sealed off. The rate of pressure leakage shall be measured and shall not exceed 0.050 inch Hg per minute.

4.6.2.5 Altitude data channel accuracy. The altitude data channel shall be tested. The altitude shall be increased by gradually reducing the input pressure corresponding to the respective altitude levels specified in table II from sea level to 80,000 feet. The output shall accurately indicate the specified pressure inputs. The pressure reading corresponding to the output for increasing altitude test points shall be recorded. The recorded pressure shall be within the minimum-to-maximum pressure range. If the pressure falls outside of this range, the deviation shall be recorded. The altitude shall be decreased from 80,000 feet to sea level and the pressure corresponding to the respective altitude switching level shall be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.2.6 Indicated airspeed data channel accuracy. The indicated airspeed data channel shall be tested. The IAS pressure input corresponding to the respective IAS switching levels specified in table II shall be increased from 0 to 800 knots. The output shall accurately indicate the specified pressure inputs. The pressure rating corresponding to the output for increasing indicated airspeed test points shall be recorded. The recorded pressure shall be within the minimum-to-maximum pressure range. If the pressure falls outside of this range, the deviation shall be recorded in the column provided. The IAS shall be decreased from 800 to 0 knots and the pressure corresponding to the respective IAS switching levels shall be recorded.

4.6.2.7 Pressure transducer switch. Altitude and IAS pressure transducer switching functions shall be tested to determine exact switching characteristics of the recording set. Exact switching points of both channels shall be determined by varying input pressures and observing appropriate output indicators. Specified increments of pressure shall be added to these input pressure values which will establish vibration test pressure levels. The recording set shall then be vibrated between 5 and 1,500 Hz at a logarithmic rate. Pressure transducer switching data shall be accumulated for analysis on a tape magazine which contains no previous accumulation of data.

4.6.3 Acceleration data channel accuracy. Three methods for testing the acceleration data channel accuracy shall be provided. The method to be followed shall be a function of the particular type of test equipment available at the facility performing the test. Any one of the following three methods shall be considered valid tests, and only one need be selected for verification of the accelerometer data channel accuracy. All errors which occur can only occur up to the point where the acceleration data becomes digitized and the output of the acceleration level counter is the point at which the data becomes digital. Once the data has become digital, no further errors shall occur. The digital operation of the recorder and recording set shall be tested as specified in 4.6.6.

a. Method A. By observation of the acceleration level counter output test points, a specific level corresponding to a particular voltage input shall be determined. The counter output test point shall be suitably displayed and the level at which coincidence is occurring in the comparator shall be easily defined. For this method, an accelerometer simulator, as approved by the procuring activity, shall be introduced into the system in place of the actual accelerometer and the voltages which cause coincidence at each of the specified levels shall be recorded. The accelerometer shall then be electrically torqued to determine the error contributed by the accelerometer at each of the specified recorded levels. (See table V.) The difference between the torquing currents and actual current output of the accelerometer shall then be related to acceleration error to establish total acceleration error at each of the levels.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE V. Accelerometer Data Channel Accuracy Data (-3 to +9G Range)

Level	Voltage	Voltage	Nominal G	Actual G
1	0.703	____	-1.80	____
2	1.171	____	-1.00	____
3	1.406	____	-0.60	____
4	1.639	____	-0.20	____
5	1.874	____	0.20	____
6	2.108	____	0.60	____
A	2.225	____	(0.80)	____
B	2.343	____	(1.00)	____
C	2.460	____	(1.20)	____
7	2.675	____	1.40	____
8	2.811	____	1.80	____
9	3.045	____	2.20	____
10	3.279	____	2.60	____
11	3.514	____	3.00	____
12	3.982	____	3.80	____
13	4.451	____	4.60	____
14	4.919	____	5.40	____
15	5.622	____	6.60	____
16	6.325	____	7.80	____
	7.028	____	9.00	____

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

b. Method B. As specified in method A, the particular voltage inputs shall be established and recorded. If a rate table of sufficient accuracy is available, a static acceleration shall be applied to the accelerometer to obtain those specified voltage outputs recorded and the applied accelerations read directly in gravitational units from the rate table display device. The difference between the actual acceleration input and that which should correspond to the particular levels shall represent the total acceleration errors.

c. Method C. If the rate table permits a rapid enough control for static acceleration input adjustment, the accelerometer output may be applied directly to the input of the system. In this method, each time an acceleration level change is indicated, the static acceleration input shall be read and recorded. The deviation between the actual recorded and the desired acceleration inputs shall represent the total acceleration errors at each of the levels. The acceleration level counter output test points shall be displayed by means of a suitable binary display device.

4.6.3.1 Accelerometer data channel accuracy (detailed). The accelerometer data channel accuracy (detailed) test shall be conducted as follows:

4.6.3.1.1 The accelerometer simulator shall be connected to the input of the recording set. The power supply input shall be adjusted while the binary display is observed. Each time an acceleration level is crossed, the voltage as read on the digital voltmeter shall be recorded. Table V shall be used for recording the voltabe levels corresponding to each of the specified level transition points. While adjusting the power supply, care shall be exercised to obtain the exact particular voltage corresponding to the particular acceleration level. The level to which the input voltage corresponds may be established by observing the binary display. Each of the corresponding voltage levels shall be recorded. The regulated supply shall be adjusted to each of the voltage levels as determined and recorded in the first section of this method. The supply shall be adjusted to read the voltage levels on a digital voltmeter. At each of the settings, readings shall be recorded on a second digital voltmeter. The nominal readings on the second voltmeter shall also be recorded.

4.6.3.1.2 As specified in 4.6.3.1.1, the voltage levels corresponding to the acceleration levels shall be established and recorded. A static acceleration shall be applied to the accelerometer to obtain the recorded voltages across a 50,000 ohm, ±0.01 percent, load resistor. The corresponding acceleration inputs shall be read and recorded. A column shall be provided for the nominal accelerations specified.

4.6.3.1.3 The accelerometer output shall be applied directly to the recording set input with static acceleration applied to the accelerometer. As each level of acceleration is reached (determined by the binary display of the acceleration

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the currentversion before use.

level counter output test point), the acceleration input as read from the rate table shall be recorded. A column shall be provided for the nominal accelerations specified.

4.6.4 Reference time data channel. A maximum of 60 counts shall be available in a 10-minute period from the reference time data channel. Each count shall be equal to 10 seconds. When the recording set power switch is applied, the 6-bit output from the reference time channel shall count to 10 minutes ±6 seconds and reset. This cycle shall be continually repeated while power is applied to the recording set. The average of two time-interval readings shall be the final recorded value.

4.6.5 Recording set frequency response. For this test, the recording set shall be connected with a sinusoidal function generator substituted for the regulated power supply. Prior to application of a 12-Hz sinusoidal voltage, both computer counters and playback indicators shall be reset. The sinusoidal input shall be applied with peaks of 2.037 and 2.644V for approximately 1 second. If a means of counting the number of cycles is applied, upon playback the number of peaks counted in each of the two words shall coincide with the number of cycles applied. This test shall demonstrate the ability of the recording set to respond to an input frequency of 12 Hz without loss of data. If desired, the input frequency may be increased to 32 Hz since the peak acceleration words at that frequency have the capacity to store 32 counts.

4.6.6 Recording set operation. The recording set tape magazine shall be mounted in position on the recorder. Electrical continuity between the recording set and tape magazine shall be established by means of Cannon electric connector DBH-25-002, or equivalent, which will establish contact when the magazine is mounted during operation. Known inputs of test data shall be inserted in a prescribed format, processed by the recording set, and stored on magnetic tape for subsequent analysis and comparison. During playback, a suitable binary indicating device shall provide an instantaneous representation of taped data. Tape magazines shall be preerased prior to any operational testing. A functional description of each test system control shall be provided.

4.6.6.1 Recording set position error. The recording set shall be placed on a test bench in its normal operating position and rotated 90° at a time until a total of six test point positions have been measured. The positions shall be as follows:

a. Base down

b. Base to the right

c. Base up

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:29Z
Check the source to verify that this is the current version before use.

d. Base to the left

e. Front down

f. Front up.

4.6.6.1.1 For each of the test points specified in 4.6.6.1, the following transducer inputs shall be used:

Altitude (In. Hg Abs)	IAS (In. Hg Diff)	Acceleration (g) Range
28.13	0	-1.00
27.22	1.35	-1.00
14.36	7.53	1.80
13.15	9.26	1.80
2.72	29.95	6.60
1.51	31.68	6.60

4.6.6.1.2 The difference between the readings obtained at any test point and the reading obtained at normal position shall not be large enough to cause the recording set to exceed the 2 percent of full-scale tolerance.

4.6.6.1.3 The tests specified in 4.6.6.1 through 4.6.6.1.2 shall be repeated at operating voltages of 21 and 29V dc.

4.6.7 Vibration (with isolators). The recording set shall be securely mounted to a vibration table and vibrated in the three planes shown on figure 10. All electrical and pneumatic lines shall be securely fastened to the recording set. The recording set shall be vibrated under room ambient conditions in accordance with method 514, equipment category (b), curve D of MIL-STD-810 except as follows. This test shall be repeated at +71° and -55°C. The recording set shall not be vibrated over 1,500 Hz.

Double amplitude of 0.20 inch from 5 to 13 Hz,
±0.41g from 13 to 15 Hz, 0.036 inch from 16 to 72 Hz,
±10g from 72 to 500 Hz, 0.0002 inch from 500 to 700 Hz,
±5g from 700 to 1,000 Hz, and ±3g from 1,000 to 1,500 Hz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.7.1 Resonance survey (Z-axis). With the vibration table so energized that it moves in the Z-axis, the frequency shall be varied in accordance with MIL-STD-810, curve D, except as follows:

a. Double amplitude of 0.20 inch from 5 to 13 Hz
b. ±0.41g from 13 to 15 Hz, 0.036 inch from 16 to 72 Hz
c. ±10g from 72 to 500 Hz, 0.0002 inch from 500 to 700 Hz
d. ±5g from 700 to 1,000 Hz, and ±3g from 1,000 to 1,500 Hz.

4.6.7.1.1 During the resonance survey, inputs shall be applied and output recorded in accordance with tables II, VI, and VII. Data for tables II and VI shall be taken with a separate tape magazine. Evidence of discontinuity or erratic behavior shall be recorded. During the survey research, all resonant frequencies specified in table VII shall be recorded.

4.6.7.1.2 Resonance vibration (Z-axis). The recording set shall be vibrated at the resonant frequency or frequencies determined in 4.6.7.1 in accordance with table VII. Throughout the test, recording set inputs and outputs shall be recorded as specified in tables II and VI. Evidence of discontinuity or erratic behavior shall be recorded. Upon completion of this test, the recording set shall be inspected for evidence of mechanical failure. Any such evidence shall be recorded.

4.6.7.1.3 Cycling (Z-axis). With the recording set mounted in the horizontal direction, the vibration frequency shall be varied from 5 to 1,500 to 5 Hz in an approximate 30-minute period for the cycling time specified in table VII. The rate of frequency change shall be logarithmic. If provisions for logarithmic cycling are not available, a linear rate may be used. During the cycling test, a performance test shall be conducted in accordance with table VIII and shall be taken on the recorder magazine. Upon completion of this test, any evidence of malfunction shall be recorded.

4.6.7.2 Resonance survey (Y-axis). With the axis of vibration along the Y-axis, the tests specified in 4.6.7.1, 4.6.7.1.1, 4.6.7.1.2, and 4.6.7.1.3 shall be repeated. Data shall be taken in accordance with table II and VI through VIII.

4.6.7.3 Resonance survey (X-axis). With the axis of vibration along the X-axis, the tests specified in 4.6.7.1, 4.6.7.1.1, 4.6.7.1.2, and 4.6.7.1.3 shall be repeated. Data shall be taken in accordance with tables II and VI through VIII.

4.6.7.4 Vibration (with isolators removed). The recording set isolators shall be blocked up so that there will be no vibration isolation. The recording set shall be operated during this test; however, no specific performance will be required. The equipment shall be operated at room ambient conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE VI. Accelerometer Data Channel Accuracy

Level	Voltage (Volts D-C)[1]	
	Max	Actual Voltage (Record)
1	0.703	
2	1.171	
3	1.406	
4	1.639	
5	1.874	
6	2.108	
A	2.225	
B	2.343	
C	2.460	
7	2.675	
8	2.811	
9	3.045	
10	3.279	
11	3.514	
12	3.982	
13	4.451	
14	4.919	
15	5.622	
16	6.325	

Note[1] : Values based on full-scale voltage of 7.028V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE VII. Mechanical Vibration Test (with isolators)

Axis Vibration					
Number of resonances	0	1	2	3	4
Resonance frequencies (in Hz)					
Vibration time per resonance 30 min per resonance					
Cycling time (hours)					

TABLE VIII. Pressure Transducer Switch Test

Function	Procedure		Freq (Hz)	Record	Playback
	Min Inches Hg	Max Inches Hg			
450 knots			5-1,500	BOF	
30,000 feet					
450 knots			1,500-5	BOF	
30,000 feet					
550 knots			5-1,500	BOF	
50,000 feet					
550 knots			1,500-5	BOF	
50,000 feet					

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.7.4.1 Resonance survey (Z-axis). With the vibration table so energized that it moves in the Z-axis, the frequency shall be varied in accordance with MIL-STD-810, method 514, curve AR, except at a double amplitude of 0.01 inch from 5 to 65 Hz, and 2g between 65 and 500 Hz. During this test, the number of resonance modes and the resonance frequency specified in table IX shall be recorded, and the recording shall not be vibrated over 500 Hz.

4.6.7.4.2 Resonance vibration (Z-axis). The recording set shall be vibrated at the resonance points determined in 4.6.7.4.1 in accordance with the amplitude and frequency specified in table VIII. The recording set shall be vibrated at each of the resonant modes for 10 minutes, and a performance check shall be conducted and recorded. The recording set shall be inspected for evidence of mechanical failure. Any such failure shall be recorded.

4.6.7.4.3 Cycling (Z-axis). The recording set shall be vibrated in the horizontal direction with the frequency ranging between 5 and 500 Hz as follows: double amplitude of 0.01 inch between 5 and 65 Hz and 2g between 65 and 500 Hz. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will be completed in approximately 15 minutes. The cycling test shall continue for two cycles. During this test, a performance check shall be conducted. The recording set shall be inspected for evidence of mechanical failure. Any such failure shall be recorded.

4.6.7.4.4 Vibration (Y-axis). With the axis of vibration along the Y-axis, the tests specified in 4.6.7.4.1 through 4.6.7.4.3 shall be repeated.

4.6.7.4.5 Vibration (X-axis). With the axis of vibration along the X-axis, the tests specified in 4.6.7.4.1 through 4.6.7.4.3 shall be repeated.

4.6.8 Low temperature and altitude. The low temperature and altitude tests shall be conducted in accordance with MIL-STD-810, method 504, and as follows:

4.6.8.1 The recording set shall be placed in the test chamber at room temperature. All required electrical and pneumatic connections shall be brought out of the test chamber.

4.6.8.1.1 Acceleration data channel accuracy. The accelerometer simulator shall be connected to the input of the recording set. The acceleration level counter output test points shall be displayed and measurements shall be taken as specified in 4.6.3.

4.6.8.1.2 Altitude and indicated airspeed data channel accuracy. With the recording set connected to an appropriate test device, altitude and indicated airspeed equivalent pressures shall be applied to the system. Output indicators shall change in accordance with input pressure variations. Measurements shall be taken as specified in 4.6.2.5 and 4.6.2.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE IX. Sequence of Vibration Test (with isolators)

Z-Axis Vibration					
Number of resonances	0	1	2	3	4
Resonance frequencies (Hz)					
Vibration time per resonance					
Cycling time (hours)					

Y-Axis Vibration					
Number of resonances	0	1	2	3	4
Resonance frequencies (Hz)					
Vibration time per resonance					
Cycling time (hours)					

X-Axis Vibration					
Number of resonances	0	1	2	3	4
Resonance frequencies (Hz)					
Vibration time per resonance					
Cycling time (hours)					

Z performance check	
Y performance check	
X performance check	

4.6.8.1.3 Recording set operation. The tape magazine shall be connected to the computer-recorder during this test. Appropriate inputs shall be applied to determine acceleration, altitude, and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and 4.6.3. Appropriate counter levels shall be checked by overflow of counter capacity and various acceleration profiles shall be generated and applied to the recording set. During playback, a suitable binary indicating device shall provide an instantaneous representation of taped data. Tape magazines shall be erased prior to any operational testing.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.8.2 Low temperature. The low temperature test shall be conducted as follows:

4.6.8.2.1 The internal temperature of the test chamber shall be reduced to -62°C and the unit allowed to stabilize. This temperature and atmospheric pressure shall be maintained for 2 hours with the recording set nonoperative. The internal temperature of the test chamber shall then be raised to -54°C and maintained until the equipment has stabilized. The pressure within the chamber shall be atmospheric. Low-voltage power (21V dc) shall be applied to the recording set and within 5 minutes of the time that power is initially applied, appropriate inputs shall be applied to determine acceleration, altitude and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and 4.6.3. The data frame shall be stored on tape and recorded through a suitable reproducer.

4.6.8.2.2 The equipment shall be turned off and the internal atmospheric pressure of the chamber reduced to 80,000 feet. The chamber temperature shall be maintained at -54°C. Upon reaching the specified altitude, the equipment high-voltage power (29V dc) shall be turned on and within 5 minutes of the time that power is initially applied, overflow of level-1 data shall be effected.

4.6.8.3 Upon completion of the tests specified in 4.6.8.2.1 and 4.6.8.2.2, a room reference run No. 3 shall be made causing overflow of level 1 and compared with reference run No. 1. This shall be done for high-voltage power input (29V dc).

4.6.9 High temperature and altitude. The high temperature and altitude tests shall be conducted in accordance with MIL-STD-810, method 504, and as follows:

4.6.9.1 With the recorder in the test chamber and the temperature adjusted to room conditions at standard atmospheric pressure, various acceleration profiles shall be generated and applied to the recording set. This shall be accomplished at standard input voltage (28V dc).

4.6.9.2 High temperature. The high temperature test shall be conducted as follows:

4.6.9.2.1 With the equipment nonoperating, the chamber temperature shall be raised to 71°C. The chamber pressure shall be atmospheric. The equipment shall be stabilized at this temperature and maintained for 16 hours. The stabilized temperature shall be recorded. High-voltage power (29V dc) shall be applied to the recording set and maintained for 4 hours. At the end of the 4-hour period, appropriate inputs shall be applied to test acceleration, altitude, and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.3. Appropriate counter levels shall be checked by overflow of counter capacity and various acceleration profiles shall be generated and applied to the recording set. The data frame shall be put on tape and recorded through the ground reproducer. The chamber temperature shall be taken every 30 minutes by means of a thermocouple and recorded.

4.6.9.2.2 With the equipment nonoperating, the chamber shall be adjusted to 36°C. After the equipment has stabilized, the chamber pressure shall be adjusted to 50,000 feet. Power (29V dc) shall be applied and the recording set operated for 4 hours. Thermocouple readings shall be taken every 30 minutes and recorded. At the end of the 4-hour period, appropriate inputs shall be applied to determine acceleration, altitude, and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and 4.6.3. Appropriate counter levels shall be checked by overflow of counter capacity and various acceleration profiles shall be generated and applied to the recording set.

4.6.9.2.3 With the equipment nonoperating, the chamber shall be adjusted to 60°C at 50,000 feet and the equipment and chamber allowed to stabilize. The recorder shall be operated at 29V d-c input power for 4 cycles, each consisting of 30 minutes of operation and 15 minutes off. Appropriate inputs shall be applied to determine acceleration, altitude, and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and 4.6.3. Appropriate counter levels shall be checked by overflow of counter capacity and various acceleration profiles shall be generated and applied to the recording set. Upon completion of this test, the power shall be turned off. The thermocouple readings shall be recorded every 10 minutes.

4.6.9.2.4 With the equipment nonoperating, the chamber temperature shall be adjusted to 20°C. After the chamber temperature has stabilized, 29V d-c power shall be applied, the chamber pressure adjusted to 80,000 feet, and the recording set operated for 4 hours. The thermocouple temperature shall be recorded every 30 minutes. At the end of 4 hours, appropriate inputs shall be applied to determine altitude and indicated airspeed data channel accuracies as specified in 4.6.2.5 and 4.6.2.6. The data frame shall be put on tape and recorded through the ground reproducer.

4.6.9.2.5 With the equipment nonoperating, the chamber shall be adjusted to 35°C at 80,000 feet. The equipment and chamber shall be allowed to stabilize. The recorder shall be operated at 29V d-c input power for 4 cycles, each cycle consisting of 30 minutes of operation and 15 minutes off. During the 30-minute on period, appropriate inputs shall be applied to determine acceleration, altitude, and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and 4.6.3. Appropriate counter levels shall be checked by overflow of counter capacity and various acceleration profiles shall be generated and applied to the recording set. The thermocouple readings shall be recorded every 10 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

4.6.9.3 With the equipment operating at normal input power (28V dc), the chamber shall be adjusted to room ambient temperature and pressure. The equipment shall be allowed to stabilize. Appropriate inputs shall be applied to test acceleration, altitude, and indicated airspeed data channel accuracies as specified in 4.6.2.5, 4.6.2.6, and 4.6.3. Appropriate counter levels shall be checked by overflow of counter capacity and various acceleration profiles shall be generated and applied to the recording set.

4.6.10 Temperature shock. A temperature shock test shall be conducted in accordance with method 503 of MIL-STD-810. Data shall be recorded.

4.6.11 Acceleration. An acceleration test shall be conducted in accordance with MIL-STD-810, method 513, procedure II, and as specified herein. Throughout the following acceleration tests, the recording set shall be mounted and tested as shown on figure 11. The recording set shall be securely mounted to the arm of the centrifuge at a minimum distance of 3 feet from the center of rotation to the geometric center of the recording set. A functional performance check shall be conducted during the acceleration tests while the recording set is subjected to 1 minute of steady acceleration.

4.6.11.1 Acceleration (direction I). The recording set shall be mounted in direction I as shown on figure 11. The radial accelerator shall be excited and 15g acceleration applied to the recording set with the power on. Appropriate counter levels shall be checked by overflow of counter capacity while the recording set is subjected to a steady acceleration of 15g for at least 1 minute. The power to the accelerator and recording set shall be turned off.

4.6.11.2 Acceleration (direction II). The recording set shall be mounted in direction II as shown on figure 11 and the test specified in 4.6.11.1 shall be repeated.

4.6.11.3 Acceleration test (direction III). The recording set shall be mounted in direction III as shown on figure 11 and the test specified in 4.6.11.1 shall be repeated.

4.6.12 Shock (part I). The recording set shall be subjected to shock tests in accordance with MIL-STD-810, method 516. This portion of the test shall be in accordance with procedure I, except as specified herein. The recording set shall be securely mounted on a shock machine and dropped from a height of 4 inches for 18 input shocks of 15g. The time duration shall be 11 ±1 milliseconds, the intensity shall be within ±10 percent when measured with a filter having a bandwidth of 0.2 to 250 Hz, and the maximum g-shock shall occur at approximately 5-1/2 milliseconds. Three shocks shall be applied in each of the six directions and in the numerical order shown on figure 12. Upon completion of this test, the recording set shall be inspected for structural failures in accordance with table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE X

	Direction of Shock						Mechanical Failure	
	1	2	3	4	5	6	YES	NO
Mounting base								
Vibration isolator								
Holding devices								
Structural failure								

4.6.12.1 Shock (part II) - crash safety. A recording set simulator shall be tested as specified in procedure III as follows:

4.6.12.1.1 The major modules shall be replaced by dummy loads which simulate the equipment in weight, location of center-of-gravity, approximate radius of gyration, and attach-point structure. The mounting base with simulated loads in place shall be securely mounted on the shock machine and subjected to the test specified in 4.6.12, except the intensity of shock shall be 30g. The mounting base, vibration isolators, and holding devices shall be examined for mechanical failure after each series of drops. Bending and distortion will be permitted, however, there shall be no structural failure of the attached joints and the dummy loads shall remain in place. The results of this test shall be recorded as specified in table X.

4.6.13 Additional environmental tests. The following environmental tests shall be conducted in accordance with the specified methods and procedures of MIL-STD-810:

a. Fungus: Method 508, procedure I

b. Salt fog: Method 509, procedure I

c. Dust: Method 510, procedure I

d. Explosive atmosphere: Method 511, procedure I.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the currentversion before use.

4.6.14 Electromagnetic interference. An electromagnetic interference test shall be conducted in accordance with MIL-STD-826 during quiescent periods and recording cycles. Test methods shall be as follows:

Test Methods

3001	Conducted Interference, 30 Hz to 14 kHz
3002	Conducted Interference, 14 kHz to 100 MHz
4001	Radiated Interference, 14 kHz to 10 GHz
5001	Conducted Susceptibility, 30 Hz to 150 kHz
6001	Radiated Susceptibility, 14 kHz to 10 GHz.

4.6.15 Reliability and longevity

4.6.15.1 Reliability. The five recording sets (see 4.4.1) shall be subjected to a reliability test in accordance with MIL-STD-781, test level F, test plan XXVI, except as specified herein, for 500 hours each which shall demonstrate a minimum acceptable MTBF of 200 hours. The recording set shall be vibrated at least 15 minutes during each hour of operation. During this test, the following performance characteristics shall be checked:

a. Acceleration data channel accuracy: Test at levels 4, 8, 12, and 16 of tables V and VI

b. Altitude data channel accuracy: Test at altitude levels of 1,000 to 2,000 feet and 50,000 to 60,000 feet as shown in table II

c. Indicated airspeed data channel accuracy: Test at indicated airspeeds of 250 to 300 knots and 500 to 550 knots as shown in table II

d. Recording set operation: Program cycling data as indicated in table XI.

4.6.15.1.1 Temperature cycling. The temperature cycling shall be determined in accordance with the standard method of temperature cycling specified in MIL-STD-781. The equipment shall be turned on and the lower test level temperature maintained for 2 hours. The upper test level temperature shall be maintained for 3 hours.

4.6.15.2 Longevity

4.6.15.2.1 Recording set. Two of the five recording sets specified in 4.6.15.1 shall continue to be tested through a longevity test in accordance with test plan XXVIII of MIL-STD-781 for an additional 500 hours each. The units shall be temperature cycled and vibrated as specified in 4.6.15.1. If no failures occur during the reliability test, the two units shall be selected at random from the

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

TABLE XI. Acceleration Level Data

Acceleration Levels	IAS	ALT	CB
1 (2^0)	0111	1011	1
2 (2^1), (2^0)	0111	1011	1
3 (2^1), (2^0)	0111	1011	1
4 (2^2), (2^1), (2^0)	0111	1011	1
5 (2^3), (2^2), (2^1), (2^0)	0111	1011	1
6 (2^4), (2^3), (2^2), (2^1), (2^0)	0111	1011	1
7 (2^4), (2^3), (2^2), (2^1), (2^0)	0111	1011	1
8 (2^3), (2^2), (2^1), (2^0)	0111	1011	1
9 (2^3), (2^2), (2^1), (2^0)	0111	1011	1
10 (2^3), (2^2), (2^1), (2^0)	0111	1011	1
11 (2^2), (2^1), (2^0)	0111	1011	1
12 (2^1), (2^0)	0111	1011	1
13 (2^1), (2^0)	0111	1011	1
14 (2^1), (2^0)	0111	1011	1
15 (2^1), (2^0)	0111	1011	1
16 (2^1), (2^0)+	0111	1011	1

Extraneous bits which may appear between adjacent frames as a result of switching one frame to another shall be disregarded.

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:29Z
Check the source to verify that this is the current version before use.

five units. If failures do occur, the two units selected shall be the ones which had the least failures. If the five units have the same number of failures, the two units shall be selected at random.

4.6.15.2.2 Tape magazine. Two tape magazines that have passed the functional tests shall be subjected to a longevity test under the temperature cycling and vibration conditions specified in 4.6.15.1 for 90,000 frames of data and 25 cycles of loop recording. (Note: The tape magazine capacity in a loop-recording cycle is approximately 4,000 frames. A frame consists of 70 bits which are recorded on the tape.) The recording rate shall be so accelerated that this test will be accomplished within a minimum amount of time. Only failures of the tape magazine shall be recorded. If three or more failures occur, the tape magazine will have failed the test. If no more than two failures occur, the tape magazine will have met the test. The tape magazines shall be subjected to an additional longevity test at room ambient conditions in accordance with test level A-1 of MIL-STD-781 in the playback mode until either 100 high-speed passes of the tape have accumulated on each magazine or significant deterioration is noted.

4.7 Inspection of preparation for delivery. The inspection and testing of preparation for delivery shall be in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Recording sets shall be individually preserved and packaged in accordance with MIL-P-116, method IIB and MIL-STD-794.

5.1.2 Level C. Recording sets shall be preserved and packaged in accordance with the level C requirements of MIL-STD-794.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Recording sets preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas type containers conforming to PPP-B-601. The gross weight of each container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:29Z
Check the source to verify that this is the currentversion before use.

5.2.2 Level B. Recording sets preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or Regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The A/A24U-10 recording set is intended for use in gathering acceleration, altitude, airspeed, and time data during flight. At the conclusion of the flight and when the aircraft is on the ground, the easily detachable tape magazine may be removed from the system and the recorded data may be extracted by a ground reproducer which will reproduce the data onto standard IBM, or equal, compatible tape reels for reduction by commercial data-processing equipment. Additional tape magazines will be procured and made available for flight use to replace filled magazines which are enroute to and from the data reduction facility. In-transit tape magazines will be conveyed within a tape transport case.

6.2 Order data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Provisions for and nature of the sensor(s), transducer, and reserve word to be recorded (see 3.4)

c. When design data is to be prepared and submitted with or prior to the test outline specified in 4.4.3.1 (see 3.4.1)

d. When a reliability assurance program will be required (see 3.4.4)

e. When a maintainability program will be required (see 3.4.6)

f. Number of tape magazines and tape transport cases required (see 3.6.2.23)

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the currentversion before use.

g. When remote switch is to be supplied (see 3.6.4)

h. When contractor is to prepare outline of tests, test procedures, and samples of methods of recording test data (see 4.4.3.1).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Acceleration peaks

6.3.1.1 Positive peak. A positive peak, or maximum, is a net positive acceleration increase from 1g of at least two blocks (three levels as defined in table I) and a subsequent net decrease of at least 2 blocks (three levels) from the highest point reached. The highest acceleration level reached is the peak that is counted.

6.3.1.2 Negative peak. A negative peak, or acceleration minimum, is a net acceleration decrease from 1g of at least two blocks (three levels as defined in table I) and a subsequent increase of at least 2 blocks (three levels) from that acceleration furthest from 1g in a negative direction. The furthest acceleration level reached is the peak that is counted.

6.3.1.3 Clarification. For clarification of definitions described in 6.3.1.1 and 6.3.1.2, refer to figure 13.

6.3.2 Cycle of frame recording. A cycle of frame recording is tape start, record, and tape stop.

6.3.3 Cycle of record and playback. A cycle of record and playback is the attachment of the magazine on the airborne recording set, filling tape capacity, removal of the magazine from the recording set, attachment of the magazine to a reproducer, and playback of recorded data.

6.4 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6610-F185

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

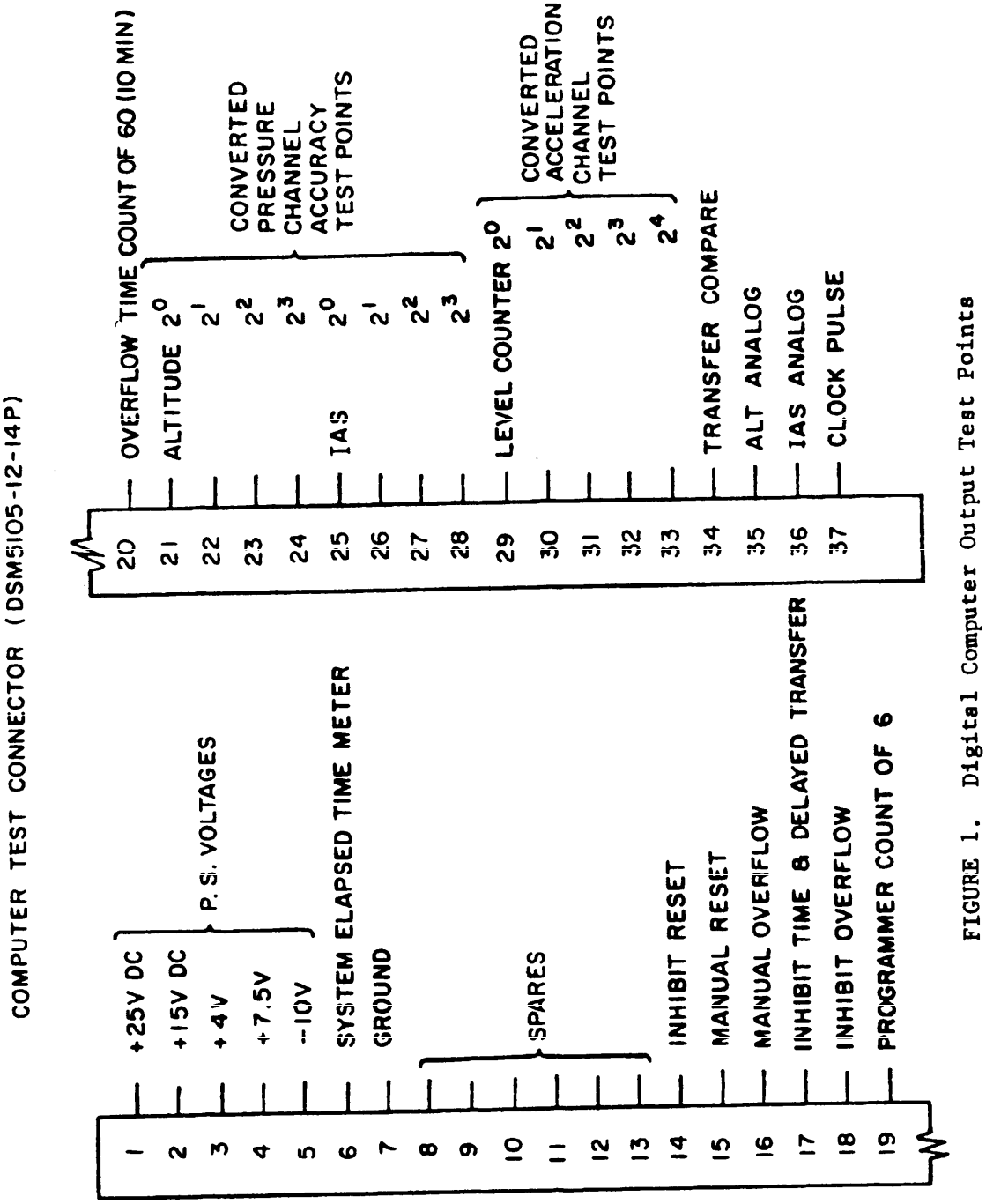


FIGURE 1. Digital Computer Output Test Points

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

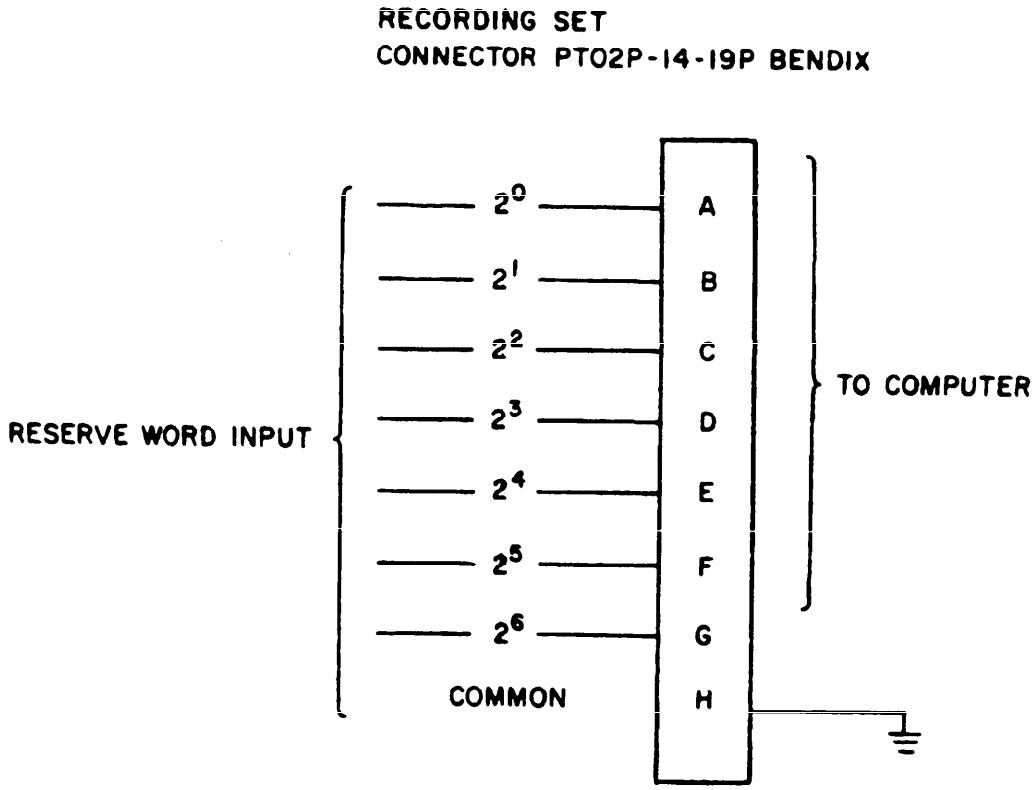


FIGURE 2. Reserve Word Interface Pin Connections

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

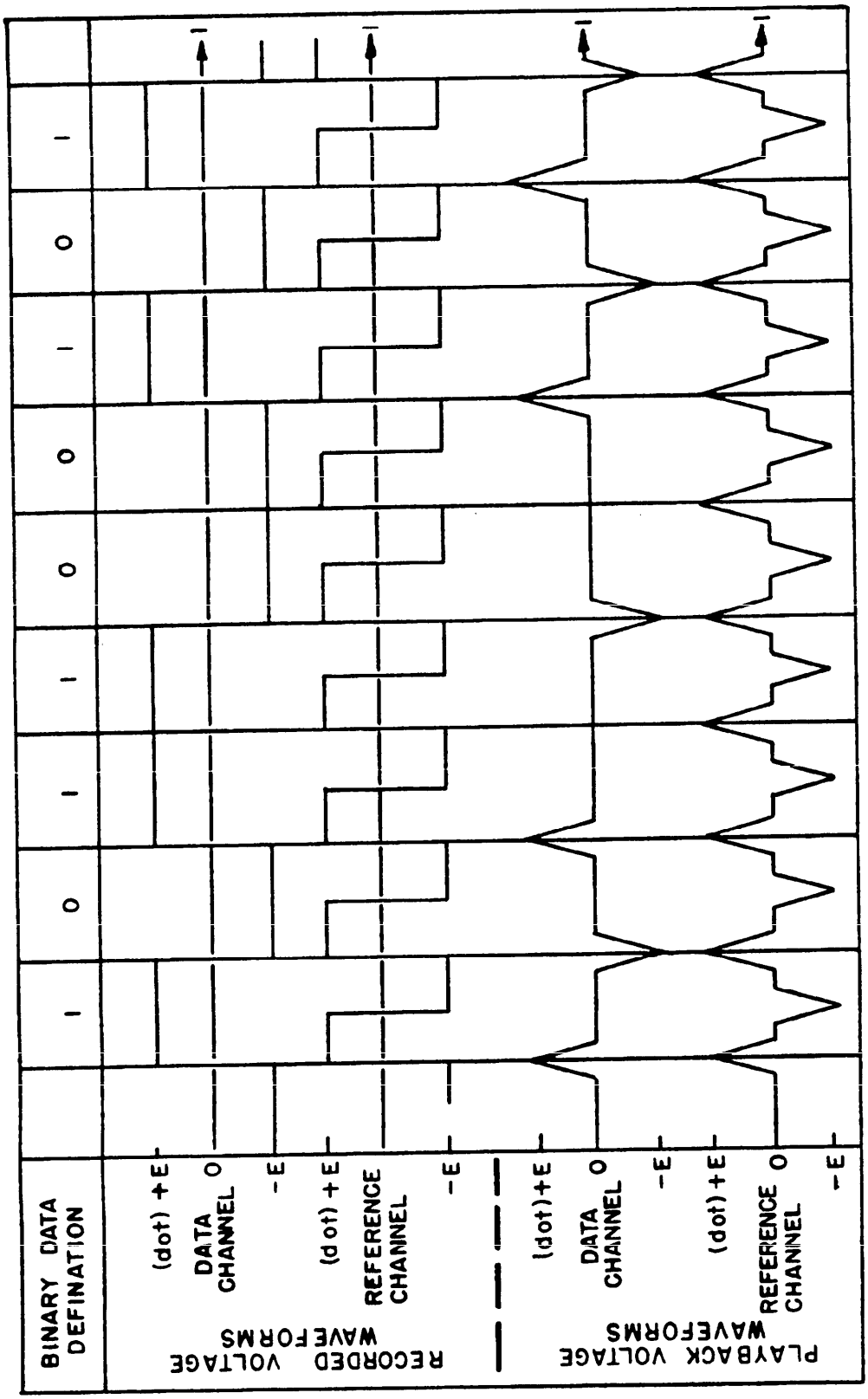


FIGURE 3. Record and Playback Voltage Waveforms

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the current version before use.

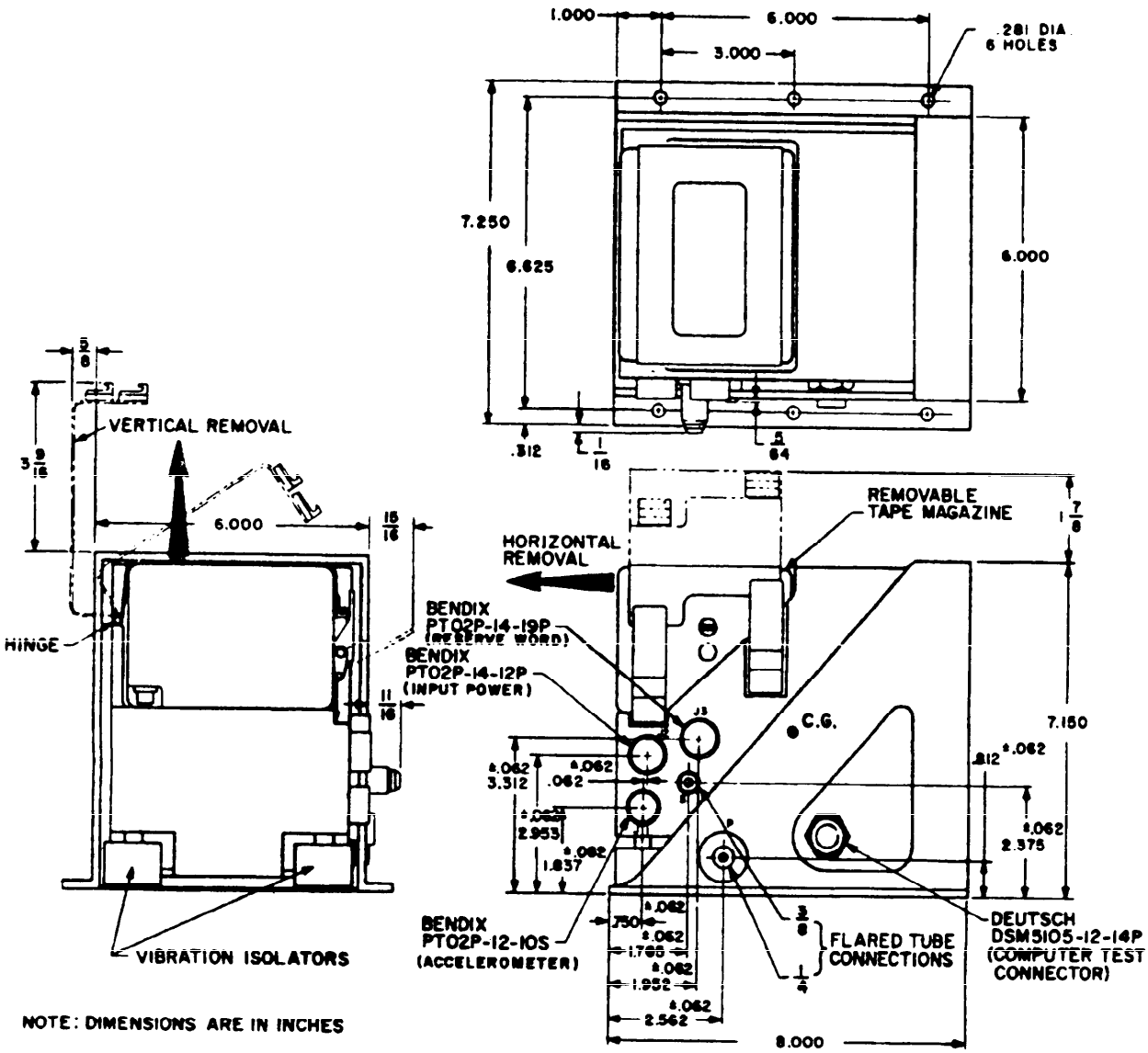


FIGURE 4. Outline Dimensions, Recorder with Tape Magazine

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:29Z
Check the source to verify that this is the current version before use.

TAPE MAGAZINE CONNECTOR
CANNON ELECTRIC DBH-25-002

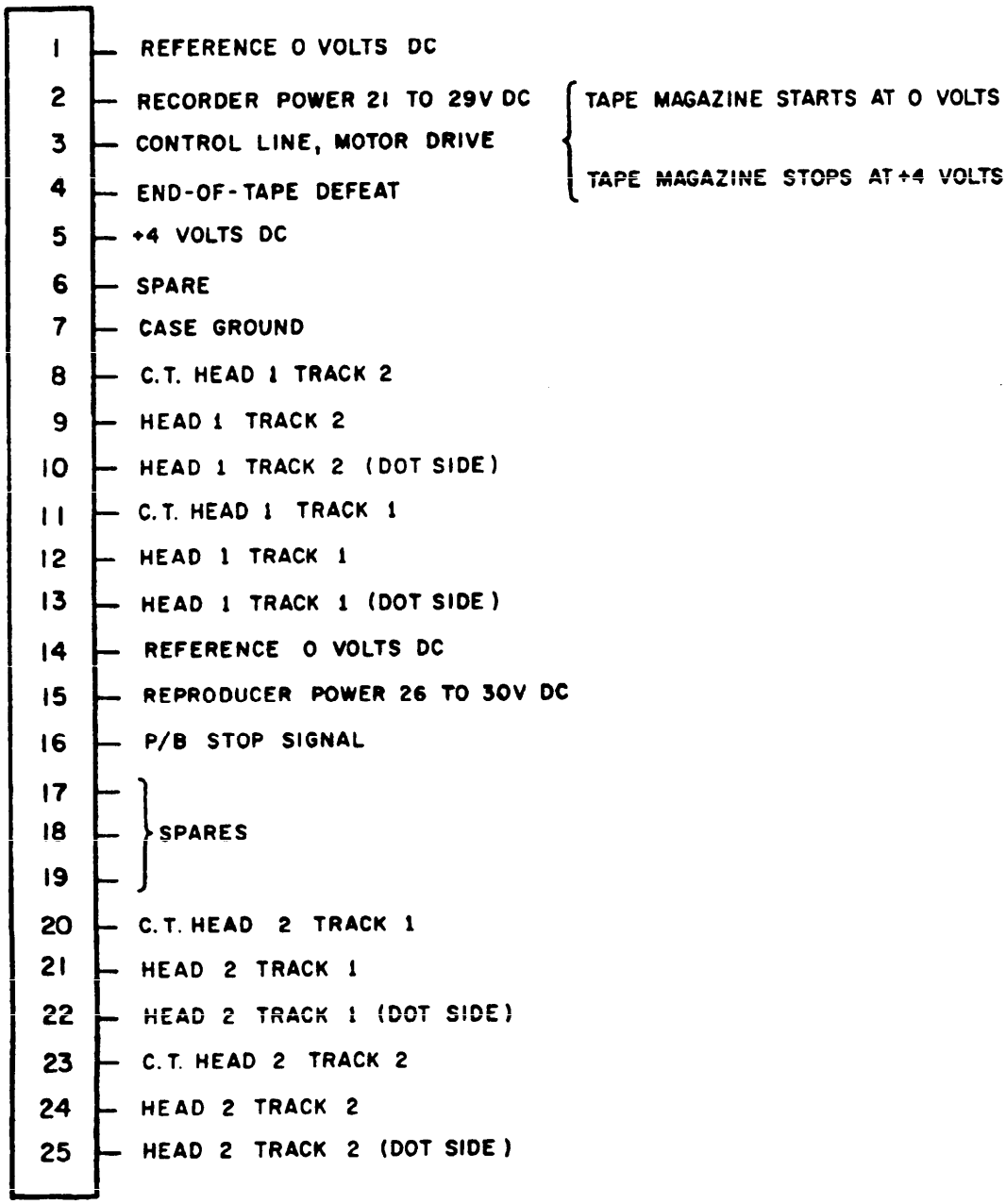


FIGURE 5. Tape Magazine, Interface Pin Connections

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

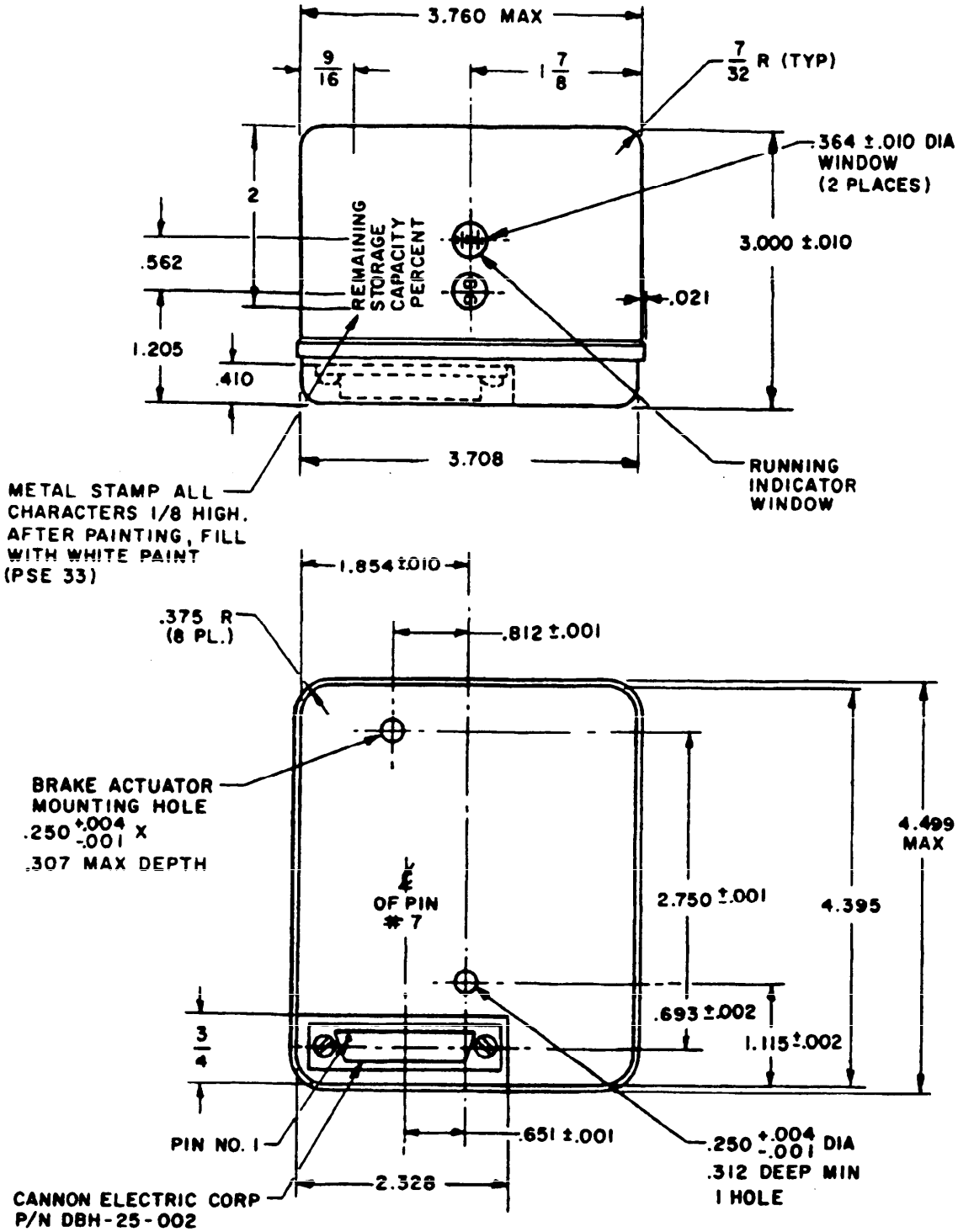
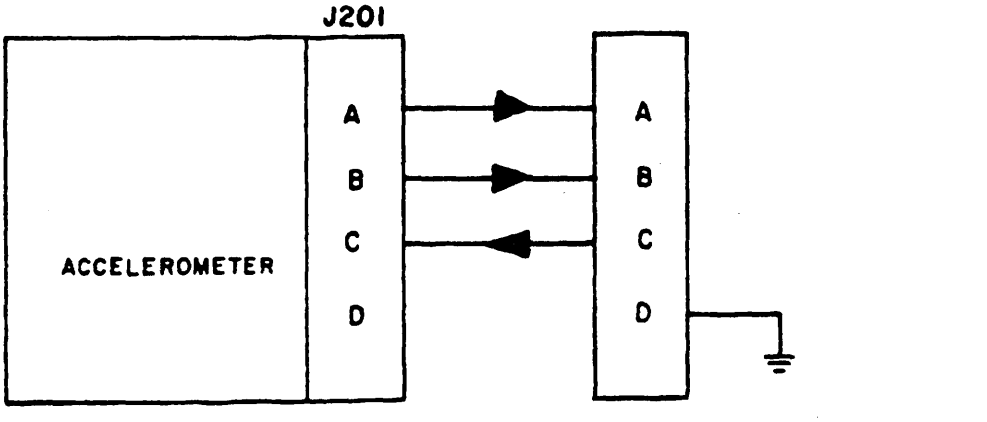


FIGURE 6. Tape Magazine Outline Dimensions

Source: https://assist.dla.mil -- Downloaded: 20260-21T12:29Z
Check the source to verify that this is the current version before use.



PINS	ACCELEROMETER FUNCTION
A	SIGNAL INPUT HIGH
B	SIGNAL AND POWER LOW
C	+28V DC HIGH
D	SPARE

ACCELEROMETER TEST PINS	
PINS	FUNCTION
A	TORQUE INPUT
B	TORQUE INPUT AND UNFILTERED OUTPUT

FIGURE 7. Accelerometer Interface

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

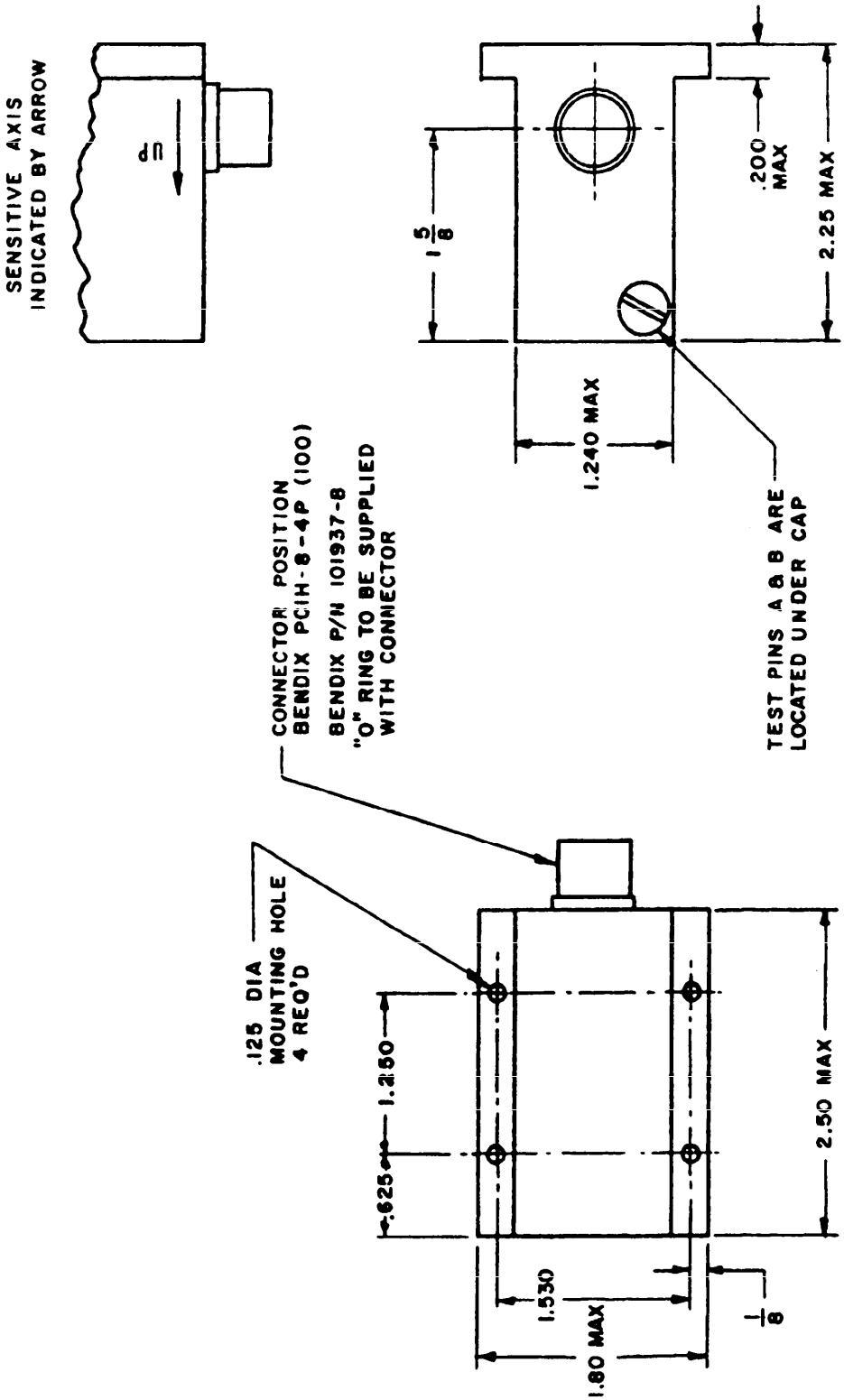


FIGURE 8. Accelerometer Outline Dimensions

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:29Z
Check the source to verify that this is the current version before use.

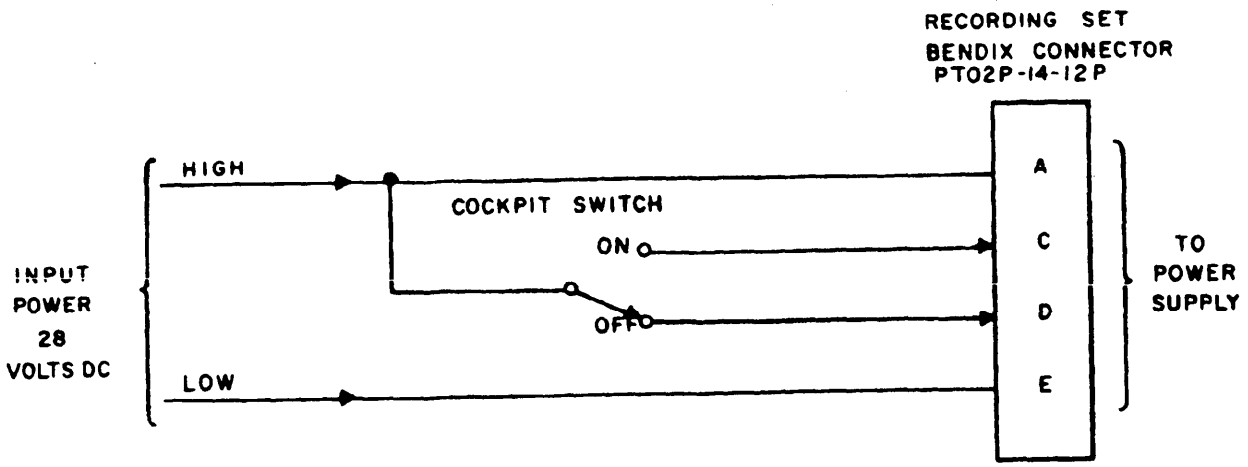
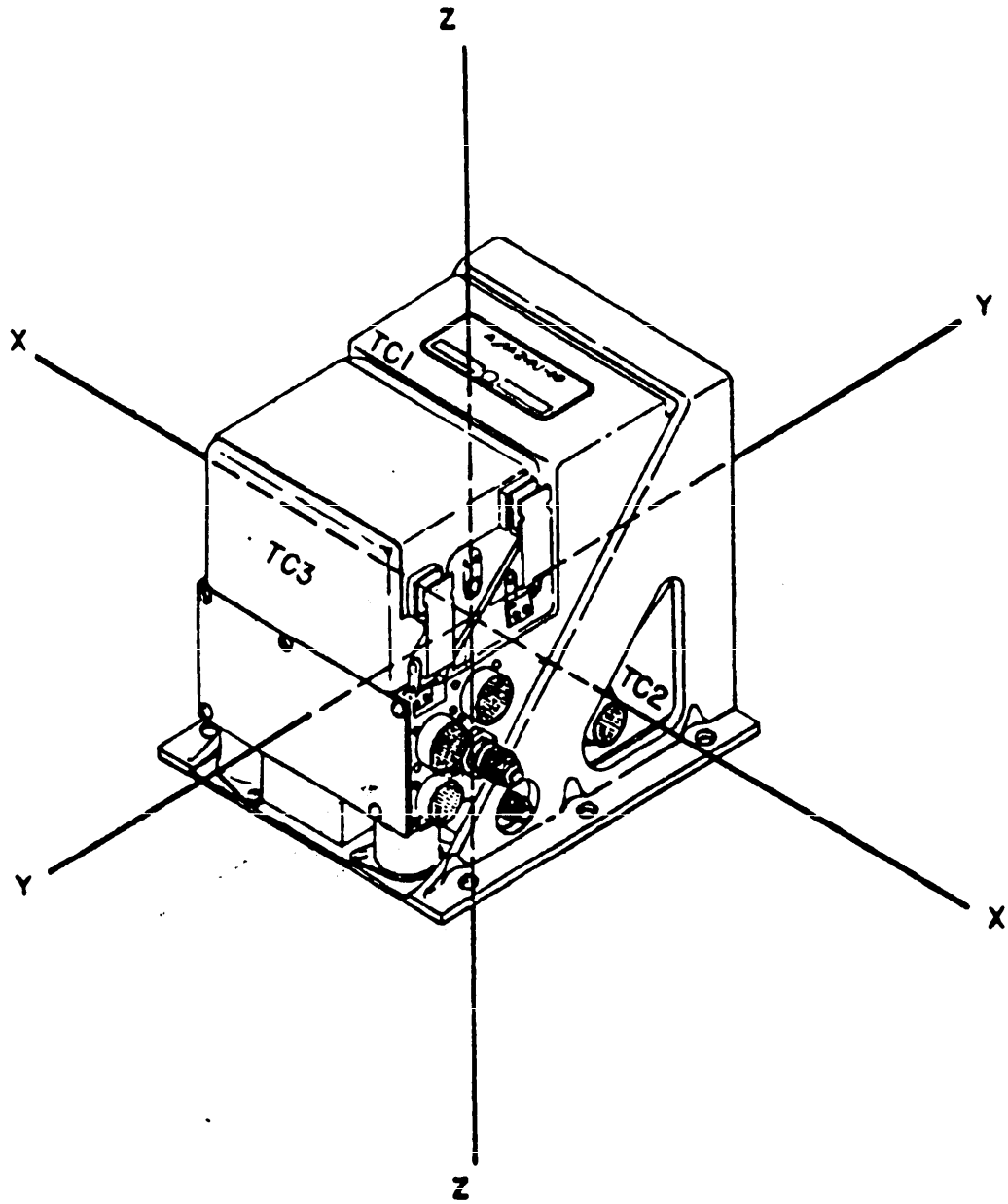


FIGURE 9. Remote Switch

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.



TC1
TC2
TC3
} LOCATION OF THERMOCOUPLES FOR RECORDING TEMPERATURE

FIGURE 10. Vibration Mounting

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

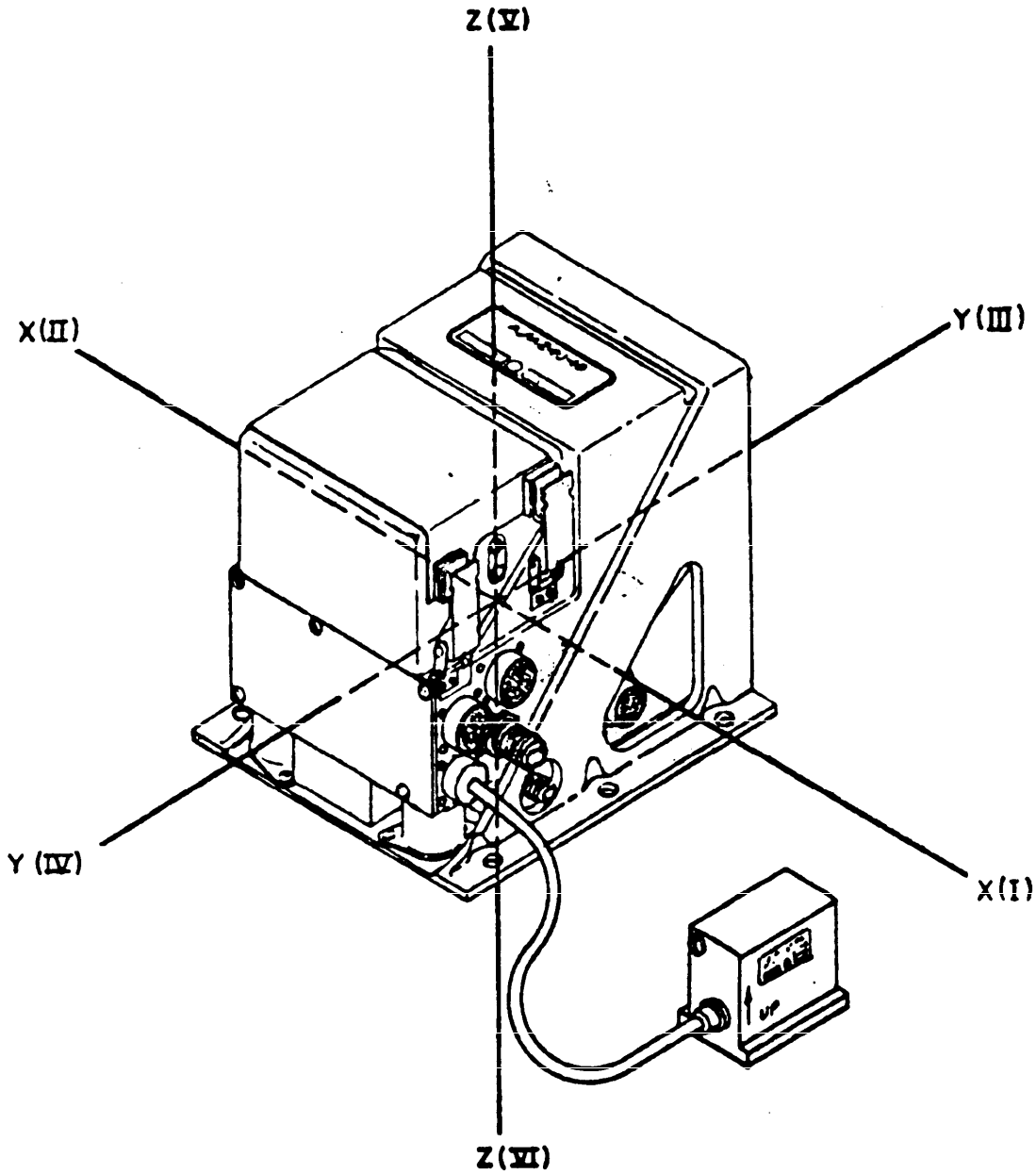


FIGURE 11. Acceleration, System Orientation

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

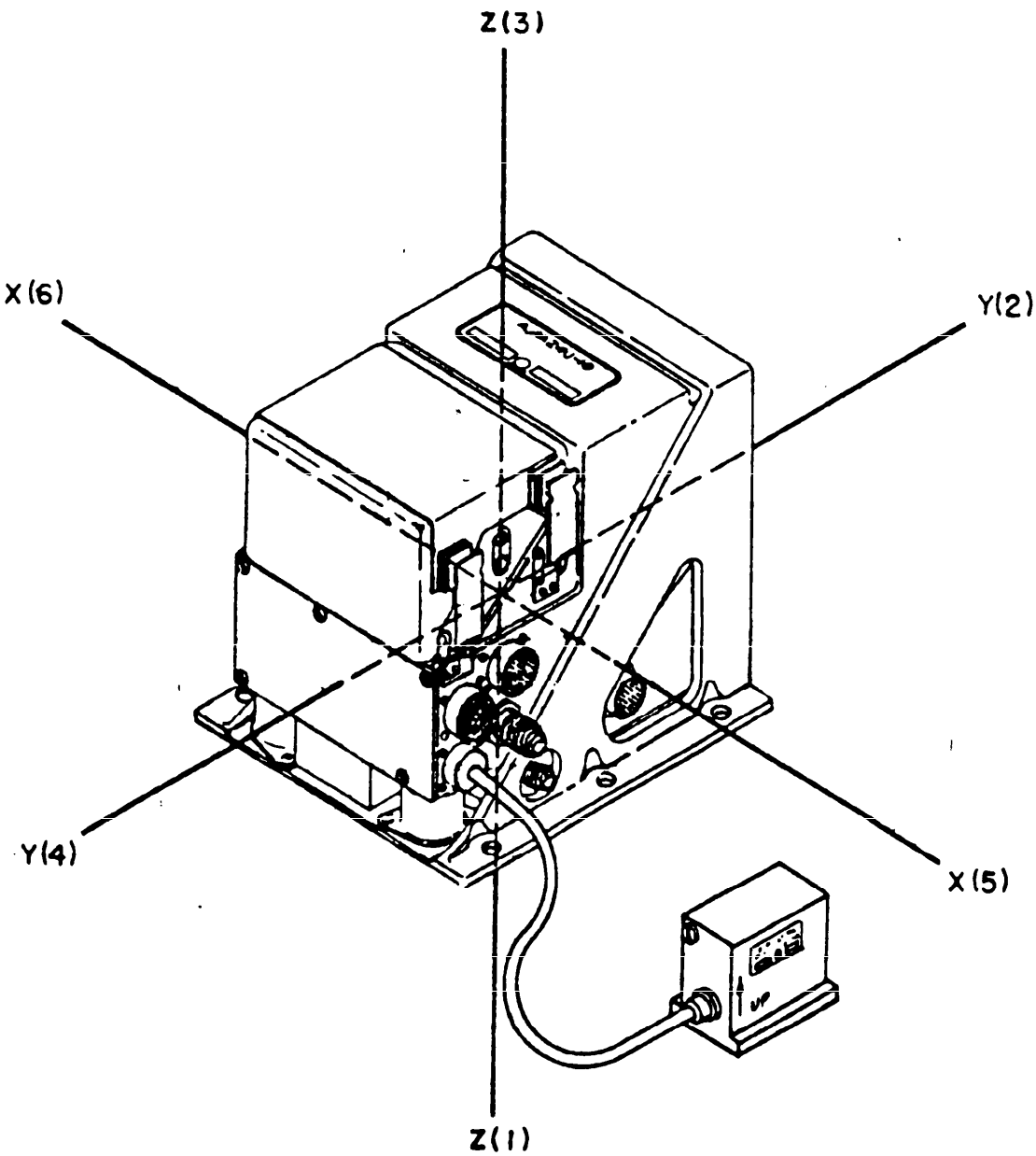


FIGURE 12. Numerical Order and Orientation of Shock

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

ACCELERATION PROFILE

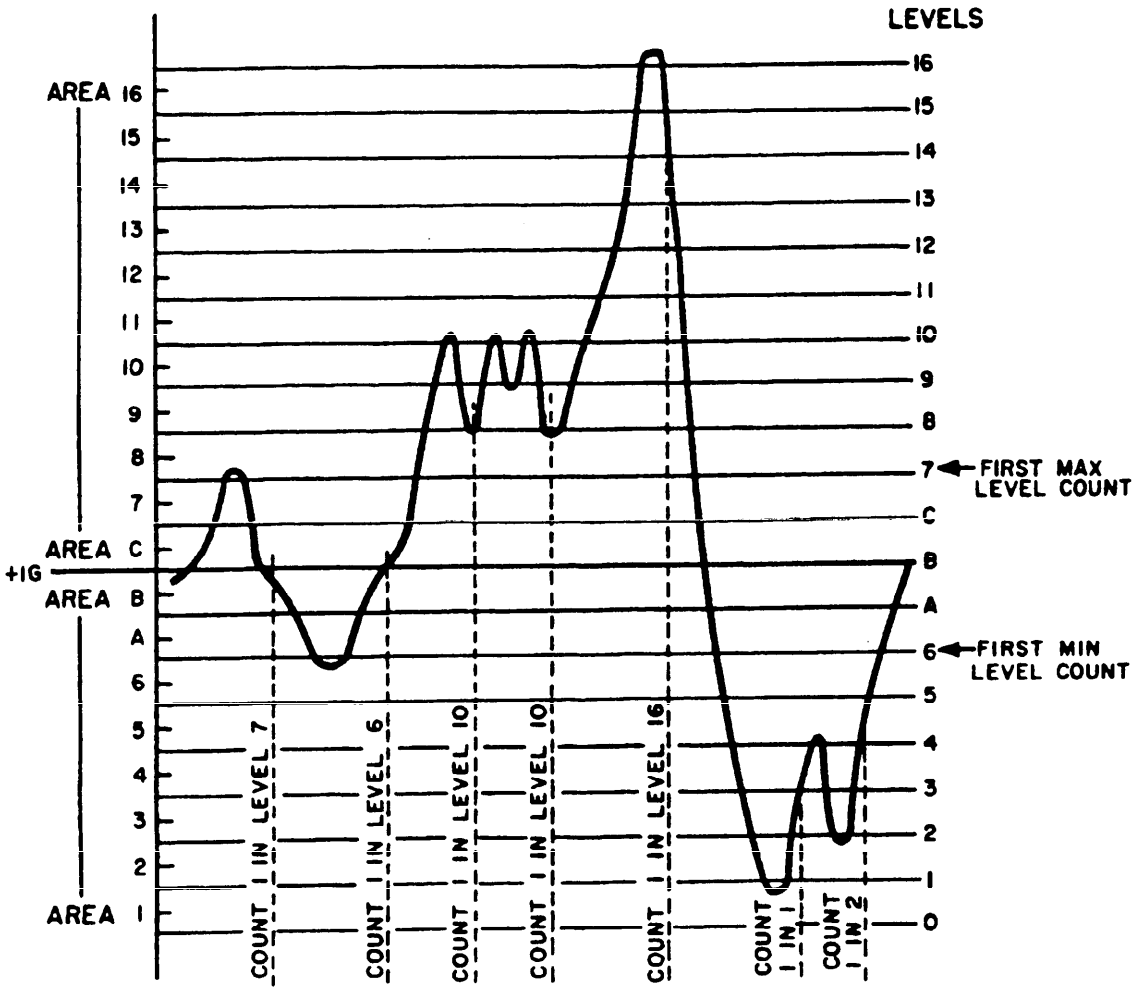


FIGURE 13. Cyclic Logic

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

C16061

Source: https://assist.dla.mil -- Downloaded: 202640-21T12:29Z
Check the source to verify that this is the currentversion before use.

FOLD

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318



FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:29Z
Check the source to verify that this is the current version before use.